

A review of Pareto pruning methods for multi-objective optimization

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ABSTRACT

Previous researchers have made impressive strides in developing algorithms and solution methodologies to address multi-objective optimization (MOO) problems in industrial engineering and associated fields. One traditional approach is to determine a Pareto optimal set that represents the trade-off between objectives. However, this approach could result in an extremely large set of solutions, making it difficult for the decision maker to identify the most promising solutions from the Pareto front. To deal with this issue, later contributors proposed alternative approaches that can autonomously draw up a shortlist of Pareto optimal solutions so that the results are more comprehensible to the decision maker. These alternative approaches are referred to as the pruning method in this review. The selection of the representative solutions in the pruning method is based on a predefined instruction, and its resolution process is mostly independent of the decision maker. To systematize studies on this aspect, we first provide the definitions of the pruning method and related terms; then, we establish a new classification of MOO methods to distinguish the pruning method from the *a priori*, *a posteriori*, and interactive methods. To facilitate readers in identifying a method that suits their interests, we further classify the pruning method by the instruction on how the representative solutions are selected, namely into the preference-based, diversity-based, efficiency-based, and problem specific methods. Ultimately, the comparative analysis of the pruning method and other MOO approaches allows us to provide insights into the current trends in the field and offer recommendations on potential research directions.

1. Introduction

This article gives an overview and a classification of multi-objective optimization (MOO) methods with an emphasis on a special class called the pruning method. In this paper, the pruning method is defined as an alternative approach that produces a focused subset of Pareto optimal solutions that are easily comprehensible to the decision maker (DM). In the pruning method, the representative solutions are selected according to a predefined instruction.

Real-world industrial engineering problems, especially strategic decision-making problems involving different stakeholders, generally entail multiple and conflicting objectives. These MOO problems are inherently difficult to address, as it is not usually possible to identify a single optimal solution that fully satisfies all the criteria. Existing review

articles typically classify MOO approaches into *a priori*, *a posteriori*, and *interactive* methods according to the resolution process and the role of the DM therein (Coello, 2000; Branke, Deb, Miettinen, & Roman, 2008; Purshouse, Deb, Mansor, Mostaghim, & Wang, 2014; Yenisey & Yagmahan, 2014; Meignan, Knust, Frayret, Pesant, & Gaud, 2015). In the *a priori* method, a multi-objective problem is converted into a single-objective problem before a single solution is obtained. The conversion process in the *a priori* methods is generally linked to the application of multi-criteria decision-analysis (MCDA) methods in determining importance weights and consolidating multiple objectives. Contrarily, the *a posteriori* method generates a Pareto optimal set, which contains a large number of solutions that are not dominated by each other. Specifically, a solution is considered Pareto optimal if an improvement on one objective cannot be made without negatively affecting at least one of other objectives. The generation of effective Pareto set usually involves

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Nomenclature			
ADA	Adaptive angle-based pruning algorithm	MDP	Minimum-diameter partitioning
AHP	Analytical hierarchy process	MSP	Maximum-split partitioning
ASA	Angle-based with specific bias parameter pruning algorithm	NOR	Non-numerical objective ranking
CHIM	Convex hull of individual minima	NSGA	Non-dominated sorting genetic algorithm
DEA	Data envelopment analysis	NSGA-II	Fast non-dominated sorting genetic algorithm
DGSOT	Dynamically growing self-organizing tree	PCA	Principal component analysis
DM	Decision maker	PROMETHEE	Preference ranking organization method for enrichment evaluations
DMM	Decision-making method	RADA	Ranked adaptive angle-based pruning algorithm
DMU	Decision-making unit	RCI	Relative cosine index
EA	Evolutionary algorithm	RCV	Relative closeness value
FCM	Fuzzy C-means clustering	RE	Relative efficiency
GD	Generational distance	REI	Relative Euclidean index
HC	Hierarchical clustering	RPB	Reference-point-based
MCDA	Multi-criteria decision analysis	SMS-EMOA	$\mathcal{S}$ metric selection evolutionary multi-objective optimization algorithm
ML	Machine learning	SOM	Self-organizing map
MOEA	Multi-objective evolutionary algorithm	SRC	Standardization based on relative clusterability
MOEA/D	Multi-objective evolutionary algorithm based on decomposition	SRD	Sum of ranking differences
MOGA	Multi-objective genetic algorithm	TOPSIS	Technique for order of preference by similarity to ideal solution
MOO	Multi-objective optimization	UBM	Utility-based measure

the use of evolutionary algorithms (EAs) due to their ability to cope with non-convex problems. Alternatively, to overcome the shortcomings of *a priori* and *a posteriori* methods, later operational researchers have developed interactive methods, which require the DM to continuously interact with an optimization algorithm to guide the search process. These interactive methods are based on the assumption that the DM's preference changes when acquiring new information about the Pareto optimal solutions.

Previous review papers have made great strides in compiling and summarizing MOO methods with an emphasis on approaches within a specific class. Some articles provided an overview of MCDA methods for *a priori* decision-making. For instance, a recent review article by Cinelli, Kadzinski, Gonzalez, and Slowinski (2020) proposed a taxonomy of the MCDA process and discussed different decision support systems for recommending relevant MCDA methods. Other similar papers that reviewed MCDA methods include Marttunen, Lienert, and Belton (2017), Watrobski, Jankowski, Ziemba, Karczmarczyk, and Ziolo (2019), Pelissari, Oliveira, Abackerli, Ben-Amor, and Assumpção (2021). Another group of review papers focused on the use of evolutionary algorithms to address MOO problems in the *a posteriori* mode. For example, Konak, Coit, and Smith (2006) compared and contrasted MOO methods based on genetic algorithms to issue guidelines on the selection of a well-suited algorithm to a particular situation. Other prominent papers focusing on specific features of EAs in MOO problems are Cheshmehgazi, Haron, and Sharifi (2015), Chugh, Sindhya, Hakanen, and Miettinen (2019). Interactive methods for MOO have also been extensively reviewed in Meignan et al. (2015), Xin et al. (2018).

Researchers in different fields have also been successful in facilitating readers by reviewing MOO methods for their specific applications. These include the applications in energy planning (Cui, Geng, Zhu, & Han, 2017; Martin-Gamboa, Iribarren, Garcia-Gusano, & Dufour, 2017; Kaya, Colak, & Terzi, 2019; Shao et al., 2020), system design (Ridha et al., 2021), environmental and system sustainability (Mansouri, Lee, & Aluko, 2015; Diaz-Balteiro, Gonzalez-Pachon, & Romero, 2017), reliability engineering (Alrabghi & Tiwari, 2015; Syan & Ramsoobag, 2019), asset management (Kabir, Sadiq, & Tesfamariam, 2014; Petchrompo & Parlikad, 2019), agricultural management (Kaim, Cord, & Volk, 2018), production planning (Laili, Li, Fang, Pham, & Zhang, 2020; Ojstersek, Brezocnik, & Buchmeister, 2020; Al-Jamimi, BinMakhashen,

Deb, & Saleh, 2021), and transportation engineering (Yannis, Kopsacheili, Dragomanovits, & Petraki, 2020).

Despite established classification schemes proposed by the aforementioned review papers, there still exist many methods that cannot be categorized as *a priori*, *a posteriori*, or interactive methods. These alternatives, referred to as the pruning method in this paper, apply pre-defined rules to reduce the number of solutions on the Pareto front. The key objective of the pruning method is to deliver an output that is more comprehensible to the DM by identifying a focused subset of Pareto optimal solutions. For example, Kulturel-Konak, Coit, and Baheranwala (2008) deployed non-numerical preference information from the DM to search for his/her preferred part of the Pareto front. Another representative example of the pruning method can be found in Zhang, Tian, and Jin (2015). The authors developed a customized EA that can identify solutions in the knee region of the Pareto front, allowing the DM to focus only on the Pareto optimal solutions with high trade-off ratios. These methods cannot be classified as *a priori* or *a posteriori* because they neither necessitate obtaining complete preference information from the DM to consolidate multiple objectives nor require the DM to finalize the decision directly from the Pareto front. Likewise, methods in this class cannot be classified as an interactive method as they do not require the DM to continuously interact with the resolution process.

Although there exist many studies that address the problem of reducing Pareto optimal solutions, it can be observed that different pruning methods are developed for different purposes. The task of searching for literature that fits a researcher's interest is even more exacting, as previous academics employed different terms to refer to their methods. Therefore, the primary objectives of this paper are to offer a precise definition of the pruning method and to systematize studies on this aspect. We examine different pruning methods found in the literature and classify them into four subclasses. To facilitate researchers in identifying methods that are suitable for their work, this paper also compare the pruning method with the MOO methods in other classes. The originality of this review paper lies in the following aspects:

- Novel classification scheme for MOO methods according to the resolution process, role of the DM, and characteristics of the final output.

- Compilation of literature and subclassification of pruning methods to facilitate readers in identifying methods that match their interests.
- Comparative analysis across *a priori*, *a posteriori*, interactive, and pruning methods to provide comprehensive guidelines on the MOO method selection.

The remainder of this paper is organized as follows. Section 2 proposes a new classification for MOO methods as well as providing precise definitions of these classes and related terms. In Sections 3–6, we discuss pruning methods in the four subclasses along with their applications. Indicators employed to assess the performance of the pruning method in the previous literature are compiled and discussed in Section 7. Section 8 presents the key findings and offers potential research avenues before the conclusions are drawn in Section 9.

2. Classification and definitions

By compiling different terms adopted by previous researchers to specify their MOO methods, this section proposes a new classification of such methods and offers formal definitions of the associated terms. Having delving into the resolution process, the characteristic of outputs, and the role of the DM, we propose classifying MOO method into four main categories: *a priori*, *a posteriori*, interactive, and pruning methods. The proposed classes are illustrated in Fig. 1 in the order of the DM's influence on the resolution process. The fundamental characteristics of MOO methods considered in this paper are summarized in Table 1.

Before discussing the characteristics of MOO methods in each class, it is important to give a precise description of the MOO problem and definitions of related concepts.

A general MOO (minimization) problem with k objectives can be described as:

$$\begin{aligned} &\text{minimize } \{f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x})\} \\ &\text{subject to:} \\ &\quad \mathbf{x} \in X \end{aligned} \quad (1)$$

where $f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x})$ are $k(\geq 2)$ objectives to be minimized, $\mathbf{x}$ is the decision vector, and X is the set of feasible solutions.

The definitions of the terms related to the MOO problem are given as follows:

Definition 2.1. (Pareto dominance) A decision vector $\mathbf{a}$ ($\mathbf{a} \in X$) is said to Pareto dominate a decision vector $\mathbf{b}$ ($\mathbf{b} \in X$) if and only if:

$$f_i(\mathbf{a}) \leq f_i(\mathbf{b}) \text{ for all } i \in \{1, 2, \dots, k\} \text{ and } f_i(\mathbf{a}) < f_i(\mathbf{b}) \text{ for at least one } i \in \{1, 2, \dots, k\} \quad (2)$$

Table 1
Features of MOO methods.

MOO class	Resolution process	Output	Role of DM
<i>A priori</i>	A multi-objective problem is converted into a single-objective problem before the optimization process takes place.	Single solution	Provide either (i) complete preference information or (ii) precise instructions on how to identify a single representative objective function.
<i>A posteriori</i>	Multiple objectives are concurrently optimized to obtain a set of non-dominated solutions.	Pareto optimal set	Finalize the solution directly from the Pareto optimal set.
Interactive	An optimal solution or a set of optimal solutions are repeatedly generated according to the feedback from the DM.	Guided solution(s)	Continuously evaluate solutions generated by EA and update preference information.
Pruning	Pruning instructions are applied before, during, or after the optimization process to reduce the number of Pareto optimal solutions.	Pruned set (reduced Pareto optimal set)	Specify pruning instructions and finalize the solution from the pruned set.

Definition 2.2. (Pareto optimal solution) A solution $\mathbf{x} \in X$ is a Pareto optimal solution (also called non-dominated solution) if there is no $\mathbf{x}' \in X$ that Pareto dominates $\mathbf{x}$.

Definition 2.3. (Pareto optimal set and Pareto front) The Pareto optimal set is a set that contains all (or as large as possible) Pareto optimal solutions. The set of objective values that correspond to the solutions in the Pareto optimal set is illustrated on a Pareto front (also called Pareto frontier).

The remainder of this section will briefly discuss the elementary characteristics of each category in our proposed classification.

2.1. *A priori* method

In the acknowledged reviews on MOO (Marler & Arora, 2004; Branke et al., 2008; Yenisey & Yagmahan, 2014; Kaim et al., 2018), the *a priori* method is used to refer to an approach that requires the DM to articulate his/her preference information to consolidate multiple objectives before

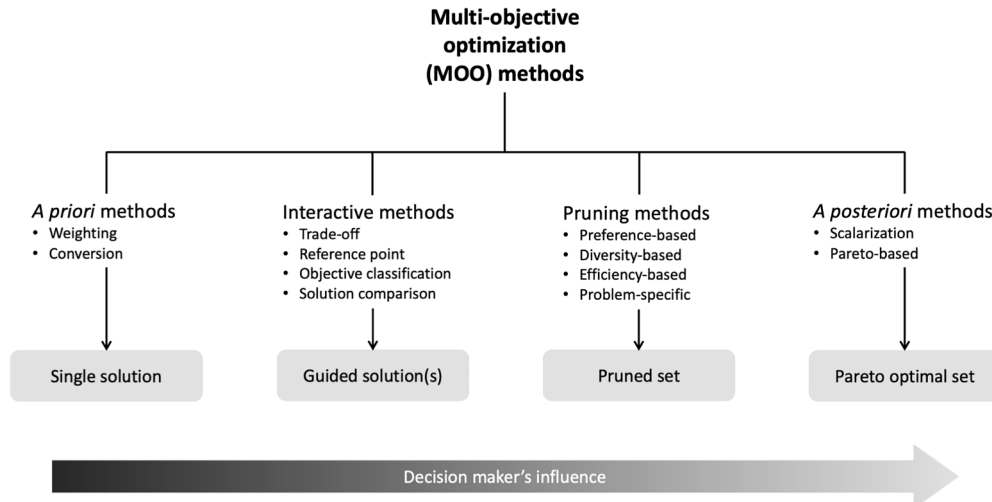


Fig. 1. Classification of MOO methods.

the optimization process takes place. The preference information can be specified in terms of the importance weight or the expected goal for each of the objectives. In other words, a multi-objective problem is transformed into a single-objective problem before it is solved by a single-objective optimization algorithm for a single optimal solution. The *a priori* method is based on the fundamental assumption that the DM possesses sufficient prior knowledge about the problem that he/she can quantify the relative importance of multiple objectives and derive a single representative objective to be optimized.

It can be observed from the existing reviews (Wang, Jing, Zhang, & Zhao, 2009; Petchrompo & Parlikad, 2019; Syan & Ramsoobag, 2019) that the *a priori* methods generally involve two major tasks: (i) determining appropriate importance weights for different objectives and (ii) converting multiple objectives into a single objective. These two tasks are referred to as weighting and conversion processes respectively. In the weighting process, the importance weights can be determined either subjectively using MCDA (Edwards, 1977; Jia, Fischer, & Dyer, 1998; Wang, Jing, Zhang, Shi, & Zhang, 2008; Ishizaka & Labib, 2011) or objectively for certain problems (Doostparast, Kolahan, & Doostparast, 2014; Khorshidi, Gunawan, & Ibrahim, 2016; Nakousi, Pascual, Anani, Kristjanpoller, & Lillo, 2018; Petchrompo, Li, Erguido, Riches, & Parlikad, 2020). On the other hand, the conversion process is generally linked to the application of the MCDA combining methods (e.g. weighted-sum and weighted product methods), the ϵ -constraint method, and the goal programming method (Lee & Kim, 2000; Marler & Arora, 2004; Mavrotas, 2009). Researchers interested in *a priori* methods are suggested to refer to Ho, Xu, and Dey (2010), Soltani, Hewage, Reza, and Sadiq (2015), Marttunen et al. (2017), Watrobski et al. (2019), Cinelli et al. (2020).

According to the discussion above, the *a priori* method can be formally defined as:

Definition 2.4. (*a priori* method) An *a priori* method is a method whereby multiple objectives are converted into a single objective before the optimization problem is solved for a single optimal solution.

2.2. A posteriori method

In many cases, the DM may neither possess any prior knowledge about the problem domain nor be able to provide any instructions to consolidate multiple objectives before the optimization process. To address this problem, previous researchers have developed a *a posteriori* methods whereby the Pareto optimal set is generated and presented to the DM before he/she can finalize the solution therefrom. This method is also known as the generate-first-choose-later approach and the method for generating Pareto optimal solutions (Marler & Arora, 2004; Branke et al., 2008). The *a posteriori* method is based primarily on the assumption that the DM is capable of choosing one final solution from a large Pareto optimal set.

The *a posteriori* methods are usually classified according to how the solutions are generated (Cui et al., 2017; Kaim et al., 2018; Gunantara, 2018). Two typical classes mentioned in existing literature are the scalarization method and the Pareto-based method. The scalarization method deploys the objective conversion techniques to convert multiple objectives into a single-objective scalar function (Miettinen & Mäkelä, 2002). The optimization problem is resolved multiple times to obtain a Pareto front (Karimi & Jadid, 2019). The Pareto-based method, on the other hand, uses the Pareto dominance concept as described in Eq. (2) to guide the search process. In this mode, multiple solutions are concurrently generated to obtain the Pareto front. Existing studies focusing on such methods generally adopted multi-objective evolutionary algorithms (MOEAs) as their solution methods. Readers interested in more recent algorithms or application-specific MOEAs applied in the *a posteriori* mode are suggested to refer to Yenisey and Yagmahan (2014), Mansouri et al. (2015), Cui et al. (2017), Gunantara (2018), Kaim et al. (2018), Chugh et al. (2019), Al-Jamimi et al. (2021), Ridha et al. (2021).

According to the scope given in the literature, the *a posteriori* method can be defined as:

Definition 2.5. (*a posteriori* method) An *a posteriori* method is a method that concurrently optimizes multiple objectives and populates the Pareto optimal set before the DM can choose one solution therefrom.

2.3. Interactive method

In an interactive method, the solutions are iteratively improved through the cooperative efforts of the optimization algorithm and the DM (Brintrup, Ramsden, Takagi, & Tiwari, 2008; Barbat, Greco, Kadzinski, & Slowinski, 2018). Interactive MOO methods generally involve two major steps. In the first step, a single solution or a small number of Pareto optimal solutions are randomly generated and presented to the DM. In the second step, the DM then expresses his/her preference towards the generated solutions. This preference information is then used to guide the search algorithms in the next iteration. The process is repeated until either a pre-set termination criterion is met or the DM is satisfied with the solution. There are two major assumptions that underpin the interactive method. First, the preference of the DM changes gradually when he/she is exposed to different solutions. Second, the generated solutions eventually converge to the DM's preferred part of the Pareto front.

Previous scholars usually classified interactive methods for MOO according to the type information extracted from the DM's feedback. Miettinen, Ruiz, and Wierzbicki (2008), Meignan et al. (2015) identified three primary categories of interactive methods: trade-off (Branke, Kaussler, & Schmeck, 2001; Lwin, Qu, & Kendall, 2014; Ji, Luo, & Peng, 2019), reference point Deb and Sundar (2006), Lopez-Jaimes and Coello Coello (2014), Barba-Gonzalez, Nebro, Garcia-Nieto, and Aldana-Montes (2020), and objective classification (Miettinen & Makela, 2000; Miettinen & Makela, 2006; Aittokoski & Miettinen, 2008). More recent techniques allowed the DM to compare and express their preference directly on some representative solutions (Branke, Greco, Slowinski, & Zielniewicz, 2015; Branke, Corrente, Greco, Slowinski, & Zielniewicz, 2016). Hence, later researchers (Xin et al., 2018) also proposed a new class, solution comparison, to accommodate these methods. A comprehensive review on the state-of-the-art interactive methods for MOO can be found in Branke et al. (2008), Bechikh, Kes-sentini, Said, and Ghedira (2015), Meignan et al. (2015), Xin et al. (2018).

According to the previous reviews on this subject, the precise definition of the interactive method is given as follows:

Definition 2.6. (*interactive* method) In an interactive multi-objective optimization method, the DM's preference is progressively elicited in the resolution process to guide the algorithm's search towards his/her regions of interest on the Pareto front.

2.4. Pruning method

To cope with a large number of solutions obtained from the *a posteriori* method, previous researchers proposed alternative tools that produce a focused Pareto optimal subset, with a primary objective to deliver the DM with the results that are more comprehensible. These methods apply a set of predefined rules to filter out or prune the solutions that are unnecessary at the primary decision-making stage. In this review, these methods are referred to as the Pareto pruning method or, in short, the pruning method.

Although both the interactive and the pruning methods aim at determining a manageable set of Pareto optimal solutions, the two are distinctly different in two main aspects. First, rather than continuously giving feedback to the algorithm during the optimization process, the DM is only asked to provide a set of rules to reduce the number of solutions in the pruning method. Second, unlike the interactive method which only takes cognizance of the DM's preference, the pruning

method allows other types of information such as the solution diversity (Taboada & Coit, 2007) or efficiency (Branke, Deb, Dierolf, & Osswald, 2004) to be incorporated in the resolution process.

The essence of the pruning method is in line with the concept of innovization, which refers to the acquisition of knowledge through the post-optimality analysis of the Pareto optimal solutions (Deb & Srinivasan, 2006). However, the pruning method considered in this review extends beyond the concept of innovization. It was evident from the previous literature that the insights to support the decision-making can be obtained not only after but also before or during the optimization process. For instance, to deal with a multi-objective system redundancy allocation problem, Kulturel-Konak et al. (2008) proposed a method in which the DM is asked to rank the objective functions in the order of their priority. This information is then applied prior to the optimization stage, allowing their multinomial tabu search algorithm to arrive at the DM's desired space on the Pareto front. In another example, Branke et al. (2004) demonstrated how a preset efficiency-based indicator is applied during the optimization process. In their study, the fast non-dominated sorting genetic algorithm (NSGA-II) with an angle-based measure embedded therein was shown to effectively identify knee regions of the Pareto front. In the remainder of this review, the terms pre-optimality, intra-optimality, and post-optimality will be used to refer to the application of the pruning method before, during, and after the optimization process respectively.

Although there exist many studies that strive for the same goal of identifying a focused Pareto optimal subset, previous researchers came up with different terms to refer to their proposed methods. For example, Taboada and Coit (2008) presented a method that coupled a multi-objective genetic algorithm (MOGA) with the non-numerical objective ranking method to identify a prioritized set of Pareto optimal solutions. The authors referred to their proposed method as the pruning method and to its output as the pruned Pareto optimal set. Jornada and Leon (2016) developed an approach to identifying a subset of robust solutions from the Pareto optimal set and termed their method the Pareto set reduction method. Thereafter, Li, Liao, and Coit (2009) applied a data envelopment analysis technique to identify a representative set of Pareto optimal solutions with high relative efficiency. Their method for determining representative solutions was referred to as the multiple objective selection optimization method. Yadollahi, Majid, and Zin (2015) applied the technique for order of preference by similarity to ideal solution (TOPSIS) to prioritize the Pareto optimal solutions generated by a MOGA and called their method the post-Pareto optimality approach. Moreover, some previous scholars also employed the terms specific to their pruning techniques to refer to their approaches. Examples include the method for clustering of the Pareto front (Zio & Bazzo, 2011), the knee-point-driven algorithm (Zhang et al., 2015), the highest hypervolume method (de la Fuente, Vega-Rodriguez, & Perez, 2018), and the game-theory-based solution selection (Cao, Coit, Hou, & Yang, 2020).

It is evident that the literature search on this aspect is an exacting task due to the different varieties of terminologies adopted by the previous researchers to refer to their methods. Furthermore, the discussion above also demonstrates that different methods are based on different assumptions and instructions for reducing the Pareto front. These aspects make it even more difficult for researchers to identify a conducive method to their interest. Hence, in this review, we propose standardizing the terms related to the pruning method by providing the following definitions:

Definition 2.7. (Pareto pruning method) A Pareto pruning method or, in short, a pruning method is a method that applies a set of predefined rules to the resolution process in order to identify a focused subset of Pareto optimal solutions.

Definition 2.8. (pruning instruction) The pruning instruction is a set of predefined rules laid down by the DM to filter out the Pareto optimal solutions that are unnecessary at the primary decision-making stage.

Definition 2.9. (representative solution) A representative solution is a Pareto optimal solution that still remains in the focused Pareto optimal subset after the pruning instruction is applied to the resolution process.

Definition 2.10. (pruned Pareto optimal set and pruned Pareto front)

The pruned Pareto optimal set or, in short, the pruned set is a set that contains all the representative solutions. The set of objective values that correspond to the solutions in the pruned set is illustrated on the pruned Pareto front.

Apart from defining these standardized terms, we also propose classifying pruning methods into four subclasses according to their pruning instructions as shown in Fig. 2. The details of these four subclasses will be explicated in Sections 3–6.

3. Preference-based pruning method

In the preference-based method, the pruning instruction is derived from the DM's preference. There are two major ways in which the preference information instruction can be employed to prune the Pareto front. First, the pruning instruction derived from the DM's partial preference information can be given before the optimization process, so that the optimization algorithm is guided towards his/her preferred region. Second, it can be applied after the Pareto optimal solutions are generated; in this case, the preference-based pruning method allows the DM to shortlist promising solutions after he/she learns about the non-dominated objective space. In this review, we classify preference-based pruning methods into four main groups as shown in Table 2. The remainder of this section will discuss these methods in detail.

3.1. Non-numerical objective ranking method

The non-numerical objective ranking (NOR) method was first proposed by Taboada, Baheranwala, Coit, and Wattanapongsakorn (2007) to deal with a multi-objective system redundancy allocation problem. The underlying assumption of this method is that the DM can express his/her preference towards the conflicting objectives to some extent. More precisely, the DM can prioritize these objectives but cannot determine exactly how much one objective is more important than another. According to this assumption, a generic NOR method, couples the information on the ranking of objectives with the *a priori* method to identify the pruned set. The detailed procedure of the NOR method based on Kulturel-Konak et al. (2008) is given in Algorithm 1.

Algorithm 1. Non-numerical objective ranking method

Input: ranking of objectives based on DM's preference, termination condition
Output: pruned set (solutions with prioritized objectives)

- 1: normalize objectives
- 2: set up weighting conditions according to ranking of objectives
- 3: **while** termination condition is not satisfied **do**
- 4: randomly generate importance weights according to ranking
- 5: sum weighted objectives to determine single-objective function
- 6: solve for optimal solution
- 7: store solution in pruned set
- 8: update termination condition
- 9: **end while**
- 10: return pruned set

The pruning instruction of NOR is based on the ranking of objectives specified by the DM. A set of weighting conditions is set out according to such instruction before the resolution process takes place. The process begins with the random generation of a feasible weight combination. A set of weights is considered feasible as long as the weights are logically consistent with the aforementioned weighting conditions and sum up to one. The weighted-sum method is applied to determine a single-objective function, which is then solved for an optimal solution. After the optimal solution is stored in the pruned set, the resolution process is repeated until the preset termination condition is satisfied. The termination condition commonly used in the existing literature is the

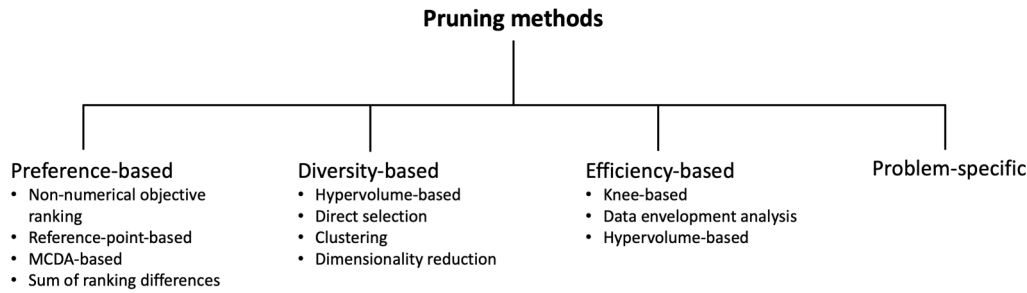


Fig. 2. Classification of pruning methods.

Table 2

Preference-based pruning methods.

Methods	Pre-optimality	Post-optimality
Non-numerical objective ranking (NOR)	✓	
Reference-point-based (RPB)	✓	
MCDA-based		✓
Sum of ranking differences (SRD)		✓

maximum number of iterations. With this method, the DM can obtain the representative solutions that yield the objective values in his/her preferred region on the Pareto front without having to specify the exact numerical weight for each objective.

The NOR method is generally coupled with a selected optimization algorithm to search for representative solutions. The NOR method was shown to work effectively with tabu search (Kulturel-Konak et al., 2008), NSGA (Taboada et al., 2007), and NSGA-II (Guo, Wong, Li, & Ren, 2013). Although it is common to have separate procedures for the preference-based pruning method and the optimization algorithm, some researchers also incorporate the NOR instruction directly into the selection step of their genetic algorithms (Zio, Baraldi, & Pedroni, 2009). In addition to its successful applications in reliability engineering as aforementioned, the NOR method was also shown to deliver pruned sets of preferred solutions in the context of production planning and scheduling (Taboada & Coit, 2008; Guo et al., 2013; Wong, Guo, & Leung, 2014).

The major advantage of NOR lies in its preference elicitation process. Specifically, it is less complicated for the DM to rank the objectives than to specify the exact importance weight for each one. This feature enables NOR to overcome the problem of obtaining oversensitive solutions in the *a priori* method in which only a small change in objective weights could lead to a completely different optimal solution.

3.2. Reference-point-based pruning method

The reference-point-based (RPB) pruning method aims at identifying the representative solutions located near one or more reference points specified by the DM. The RPB pruning method is generally based on two major assumptions. First, the DM has sufficient knowledge to supply one or more reference points prior the optimization process. Second, the preference of the DM does not change dynamically as he/she learns about new solutions during the optimization process. As long as the two assumptions hold, the optimization algorithm can deploy the supplied reference points to guide its search towards the preferred region of the Pareto front. An illustrative example of the RPB pruning process is given in Fig. 3 where black and white markers represent the solutions in the Pareto optimal set and those in the pruned set respectively.

It is evident that some existing methods in other MOO classes such as the *a priori* method (goal programming method) and the interactive method (RPB interactive method) also deploy the DM-supplied reference points in their resolution process. However, there are subtle differences between these and the RPB pruning method. In the goal

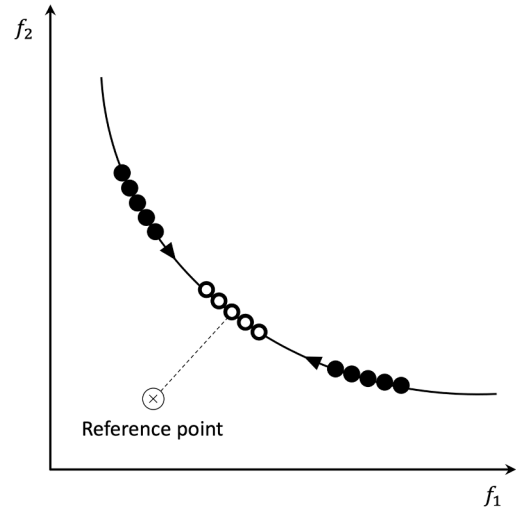


Fig. 3. RPB pruning process (adapted from Mohammadi et al. (2012)).

programming method, a representative objective function to be minimized is derived from exact penalties value for not achieving targets, leading to a single optimal solution. The RPB pruning method, on the other hand, adjusts a set of weight vectors over multiple iterations to identify the pruned set. Moreover, unlike the interactive method, the RPB pruning method does not provide the DM with an option to update the reference points during the search process.

Previous studies on the RPB pruning method usually embedded the DM-supplied reference points in their chosen MOEAs. Mohammadi, Omidvar, and Li (2012) developed the R-MEAD algorithm, which iteratively updates a set of weight vectors to identify the Pareto optimal solutions that are close to the reference points. Sudeng and Wattana-pongsakorn (2015) coupled the RPB technique with the multi-objective evolutionary algorithm based on decomposition (MOEA/D) and referred their algorithm to as PR-MOEA/D. The difference between this modified MOEA/D and the original MOEA/D is that the customized version does not use the neighborhood assignment for individual solution but only works out the closest weight vector to the updated reference point. Wang, Li, Yang, and Emmerich (2017) incorporated the DM's predefined target points into widely-used MOEAs to find solutions within the region of interest on the Pareto front. By focusing only on a limited set of solutions around the DM-supplied reference points, the aforementioned RPB pruning methods were shown to outperform normal MOEAs in terms of both the solution convergence and the computational cost.

The RPB method has been shown to arrive at robust solutions, as it can effectively filter solutions in undesirable areas of the Pareto front. This results in a faster convergence of the solutions as compared to post-optimality pruning methods. However, the effectiveness of the RPB method is highly dependent on the ideal point supplied by the DM. This means that the RPB method will be less accurate if the DM is uncertain

about the reference point, which is usually difficult to determine in a high-dimensional space.

3.3. MCDA-based pruning method

In the MCDA-based pruning method, a chosen MCDA technique is applied to reduce the number of Pareto optimal solutions obtained from the *a posteriori* method. This method assumes that the DM has the capability to evaluate and compare Pareto optimal solutions after he/she learns about the Pareto front. The MCDA-based pruning method generally leads to the ranking of solutions according to the DM's preference. The DM can then shortlist only high-ranking solutions for identifying the pruned set.

It is important to note that the MCDA-based pruning method differs from the MCDA method in the *a priori* class in that the former is used to compare the Pareto optimal solutions after the optimization process, while the latter is used to compare multiple objectives and determine appropriate weights before the optimization process.

Different MCDM techniques were used to prune the Pareto front. An interesting application of subjective MCDA methods in ranking the Pareto optimal solutions can be found in Gaytan-Iniestra and Garcia-Gutierrez (2009). In their study, the multi-objective transportation project selection problem was first solved by NSGA-II. The authors proposed using ELECTRE-III to facilitate the final selection of solutions from the Pareto optimal set. Their proposed method was shown to effectively handle the thresholds to express the preference of one solution over another. Recently, Rao and Lakshmi (2021) proposed using R-method to rank Pareto optimal solutions and identify the best compromise solution according to the importance of each objective perceived by the DM. Other subjective MCDA methods applied to prune the Pareto front in the previous literature include the preference ranking organization method for enrichment evaluations (PROMETHEE) and the analytical hierarchy process (AHP) (Wismans, Brands, Van Berkum, & Bliemer, 2014).

Other studies also used ideal and anti-ideal points to represent the DM's preference and, therefore, applied objective MCDA methods in the pruning process. A fine example of such application is the use of TOPSIS in choosing the DM's most preferred design in the reliability allocation problem of Dolatshahi-Zand and Khalili-Damghani (2015). In their research, the Pareto optimal solutions were generated by the multi-objective particle swarm optimization algorithm. To prune the large Pareto optimal set, the solutions were ranked in regard to their distance from ideal and anti-ideal solutions. Specifically, the highest ranking solution is the one located near the ideal solution and far from the anti-ideal solution according to the normalized decision matrix. Similar methods were also found to be effective in ranking the Pareto optimal solutions in other studies (Yadollahi et al., 2015; Li, He, & Zhang, 2016a; Santos, Ferreira, Flintsch, & Cerezo, 2018).

A key advantage of the MCDA-based pruning method is that there is a wide range of MCDA techniques available in the literature. As one of the most established fields in operations research, most MCDA techniques have been proved to accurately elicit the preference information from the DM in various applications. Nonetheless, different MCDA methods may lead to different pruned sets if the DM cannot provide inconsistent preference information.

3.4. Pruning method based on sum of ranking differences

The pruning method based on the sum of ranking differences (SRD) was originally developed for comparing models, methods, and analytical techniques (Heberger, 2010; Heberger & Kollar-Hunek, 2011). This method was later applied for Pareto pruning in Lourenco and Lebensztajn (2018). The application of the SRD method is based on the incorporation of multiple decision-making methods (DMMs) into the pruning process at the same time. The DMM is defined as a method that combines multiple objectives into a single representative function. The

DMMs applied in the previous literature are based on widely used MCDA methods such as SMARTS, PROMETHEE, and weighted Tchebycheff method (Lourenco & Lebensztajn, 2018; Pires, Pombo, & Lourenco, 2019). The pseudocode that represents the procedure of the SRD method adapted from Pires et al. (2019) is given in Algorithm 2.

Algorithm 2. Pruning method based on sum of ranking differences

Input: Pareto optimal set, DM-supplied importance weights, ideal point
Output: pruned set (solutions with small SRD values)

- 1: apply selected DMMs to Pareto optimal solutions
- 2: determine DMM function values for each solution based on importance weights
- 3: determine DMM function values for ideal point based on importance weights
- 4: rank DMMs for each solution according to DMM function values
- 5: rank DMMs for ideal point according to function values
- 6: calculate SRD value by sum of absolute ranking differences
- 7: identify representative solutions according to SRD values
- 8: return pruned set

There are three prerequisites for the SRD process: the Pareto optimal set, a small range of importance weights that are consistent with the DM's preference, and the ideal point. The pruning process begins with applying the selected DMMs to the Pareto optimal solutions. The DMM function values for each Pareto optimal solution and the ideal point are then calculated in accordance with the DM-supplied importance weights. The DMMs for each solution, as well as those for the ideal point, are then ranked with regard to their function values. The best value is assigned rank 1, the runner-up is assigned rank 2, and so on. The SRD value for each solution is determined by the sum of the absolute differences between the DMM rank of such solution and that of the ideal point. Finally, the representative solutions are then chosen according to the SRD value. The solutions with small SRD values are selected. This is because the SRD value closer to zero indicates the closer ranking of a solution to the ideal point.

The capability to concurrently apply multiple DMMs allows the SRD method to overcome the inconsistency in the preference information extracted from a single MCDA method. However, the SRD process can be tedious for the DM pruning as he/she needs to go over the solution evaluation process many times in order to obtain the final results.

4. Diversity-based pruning method

Diversity-based pruning methods aim at identifying a well-distributed set of Pareto optimal solutions. For a DM who cannot articulate his/her preference prior to the optimization process, it could still be a daunting task to finalize decisions even if the *a posteriori* method is adopted. This is because the Pareto front usually contains too many solutions. The rationale behind pruning methods in this class is therefore to acquaint the DM with the knowledge of the entire Pareto front using a limited subset of solutions. The pruned set of a given size is identified in such a way that it strongly represents the entire Pareto front by maximizing the solution diversity (Petchrompo, Wannakrairot, & Parlikad, 2022). Having compiled the literature on this aspect, we categorize diversity-based pruning methods into four major groups as shown in Table 3. The remainder of this section will discuss specific features of these methods in detail.

4.1. Hypervolume-based method

To identify a pruned set of diverse solutions, previous researchers

Table 3
Diversity-based pruning methods.

Methods	Intra-optimality	Post-optimality
Hypervolume-based	✓	✓
Direct selection		✓
Clustering		✓
Dimensionality reduction		✓

proposed using hypervolume — a widely-used performance indicator for MOEAs — as the pruning instruction. In its original application, the hypervolume, also known as the $\mathcal{S}$ metric, is determined by the objective space enclosed by a reference point and the Pareto front obtained from an optimization algorithm (Zitzler & Thiele, 1999). An example that demonstrates the hypervolume of a Pareto optimal set of three solutions in a bi-objective minimization problem is shown in Fig. 4. The only prerequisite for calculating hypervolume is the identification of the reference point. In the MOEA community, a slightly inferior point to the nadir point is generally used as the reference point. More details on how to identify an appropriate reference point can be found in Ishibuchi, Imada, Setoguchi, and Nojima (2018). Researchers interested in the algorithms used to approximate the hypervolume are suggested to refer to Zitzler and Künzli (2004), While, Hingston, Barone, and Huband (2006), Coello Coello, Veldhuizen, and Lamont (2007).

According to the aforementioned definition, it is apparent that both the convergence and the diversity of Pareto optimal solutions can be concurrently measured by the hypervolume. Specifically, both the spread of the solutions and their distance to the reference point contribute directly to the hypervolume value. As with the use of hypervolume in evaluating the results obtained from MOEAs, its application as a pruning indicator is also based on the rationale that a solution set is the most diverse when the space covered by its members and a selected reference point is maximized. Such instruction is usually embedded directly in an optimization algorithm to preserve the solution diversity during the resolution process.

Emmerich, Hochstrate, and Naujoks (2005) devised a method called the $\mathcal{S}$ metric selection evolutionary multi-objective optimization algorithm (SMS-EMOA), which aims at computing a well distributed set of solutions given limited size. The essence of SMS-EMOA lies in its update operation that is based on a steady-state approach. This scheme enables the algorithm to maintain the solution diversity and leads to an efficient implementation of the hypervolume-based selection strategy. Their proposed algorithm was verified using several bi-objective minimization problems. The results indicated that, in the case of the linear Pareto front, the points in the pruned Pareto set produced by SMS-EMOA are uniformly distributed. Contrarily, in the case of the convex Pareto front, more points are concentrated in the knee region around the center of the Pareto front. Although the boundary solutions are maintained, points located on flanks of the Pareto front are less dense. Extensions of SMS-EMOA can be found in Beume, Naujoks, and Emmerich (2007), Yang, Wang, Yang, Bäck, and Emmerich (2016).

For comparison purposes, Petchrompo et al. (2022) also applied the hypervolume-maximization method in the post-optimality stage. To

benchmark the hypervolume-based method against other diversity-based methods, the authors employed a generic grid search algorithm to identify a subset of Pareto optimal solutions that result in the maximum hypervolume. The findings from the application of their method to the DTLZ2 test problem were also in line with that of SMS-EMOA, as the most extreme solution for each objective is always included in the pruned set.

According to the results obtained from aforementioned studies, the hypervolume-based method has been demonstrated to consistently produce a pruned set of well-distributed solutions. Since the approximation of hypervolume values has been extensively studied in MOEA research, a diverse pruned set is guaranteed in most cases. However, the hypervolume-based pruning method does not automatically assign non-selected solutions to each of the representative solutions. This means that an additional step is required if the DM would like to further investigate neighboring solutions of his/her selected solution. Another interesting finding is that the boundary solutions on the Pareto front are always included in the pruned set produced by the hypervolume-based method. This aspect makes this technique attractive for the DM who would like to learn about extreme solutions before finalizing the decision.

4.2. Direct selection method

The simplest means of maintaining the diversity of the representative solutions is to manually select well-distributed solutions directly from the Pareto front. This method is referred to as the direct selection method in this review. In a bi-objective problem where the visualization is effective, the DM could clearly see the distribution of the Pareto front and, therefore, is capable of choosing different solutions directly therefrom. In this case, the DM can further investigate the representative solutions to discern their characteristics before finalizing the decision.

A fine example of this method can be found in Wang, Lin, Fu, Luo, and Chen (2020). After their reliability-redundancy allocation problem is solved by NSGA-II, the DM was faced with 196 solutions on the two-dimensional Pareto front. The authors proposed choosing five solutions from the frontier by first identifying two most extreme solutions (one for each objective). Other three intermediate solutions were then chosen manually at the discretion of the DM. Their proposed direct selection method was shown to provide the DM with general knowledge of the Pareto front through the comparison of the knee solution with other representative solutions distributed along the Pareto front.

It is apparent that the direct selection method is easy to implement, as it does not require any prerequisite. Nonetheless, a major drawback of this method is that its application is very limited. The selection of diverse representative solutions is based solely on the discretion of the DM. Thus, such manual selection is most effective in the two-dimensional space where both the extreme and the intermediate solutions can be directly identified from the Pareto front. The direct selection method can be problematic in a problem with more than two dimensions, as it is not possible to identify well distributed solutions without an effective visualization of the Pareto optimal solutions.

4.3. Clustering method

Another effective strategy used to preserve the diversity of the representative solution is to partition the Pareto optimal solutions into groups using a clustering method. Clustering is an unsupervised learning technique that divides data points into groups (clusters). The data is grouped in such a way that data points in the same cluster are more similar than those across different clusters. The clustering method was originally used in exploratory data mining and was later applied to various fields related to machine learning (Jain, 2010).

The clustering method was first applied to assisting the DM in multi-objective decision-making by Morse (1980). However, it was not until the study by Taboada and Coit (2007) was conducted that the concept of

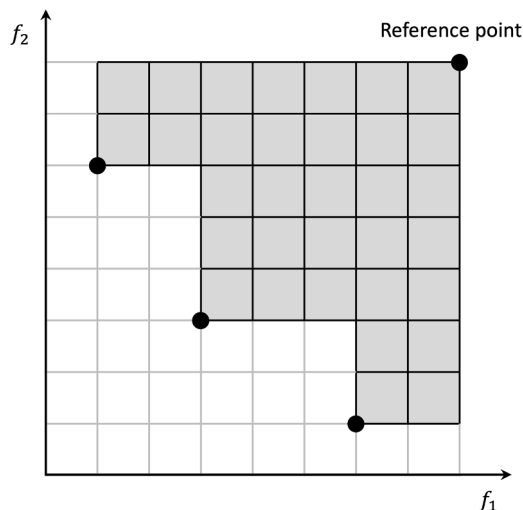


Fig. 4. Hypervolume of solutions in two dimensional space (adapted from Petchrompo (2021)).

identifying representative solutions was introduced. For its specific application in Pareto pruning, the clustering method is generally applied in the post-optimality stage to group solutions with similar objective values together. A representative solution from each group is then chosen to constitute the pruned set. The major assumption underpinning this method is that each representative solution adequately represents other solutions in its corresponding group. This means that the DM can learn about the entire Pareto front by merely examining the representative solutions at the primary decision-making stage. To illustrate, an example of an output obtained from a clustering method is given in Fig. 5, with the solution clusters differentiated by colors and the representative solutions shown in white markers.

Previous scholars have adopted various clustering methods to prune the Pareto front. In this review, we compile these methods and classify them according to the taxonomy given in Jain, Murty, and Flynn (1999) (see Fig. 6). Despite their differences in the resolution process, all clustering methods applied to Pareto pruning are performed on the basis of the distance (or similarity) between the objective function values. To ensure effective clustering results, these methods may require three additional steps: standardizing data, choosing the number of clusters, and identifying representative solutions.

Standardization is necessary for the methods in this class because multiple objectives can have markedly different scales (Brusco, 2017). Such difference could lead to unwarranted clusters obtained from the algorithm because objectives with large measurements tend to have more influence on the clustering process than do those with small measurements. By investigating related literature, we can identify two common standardizing approaches: linear normalization and standardization based on relative clusterability (SRC). The former is based solely on the range — the minimum and the maximum objective values — found in the Pareto optimal set, whereas the latter utilizes both the range and the variance of such values. The details of these standardizing methods can be found in Taboada et al. (2007), Brusco (2017).

For some clustering methods, the DM is required to specify the number of clusters prior to the clustering process. This number, in turn, corresponds to the number of representative solutions in the pruned set. Previous researchers put forward various suggestions based on both subjective and objective measures. The silhouette plot method was employed to support subjective decision-making by providing the DM with a graphical display of Pareto optimal solutions (Taboada & Coit, 2007; Taboada et al., 2007). Objective measures developed in the existing literature will be discussed in Section 7.

The identification of representative solutions is required for the methods that do not directly provide a cluster representative

corresponding to an actual solution. In this case, previous researchers suggested using the solution that is located nearest to the centroid (Taboada & Coit, 2008; Weiss, 2019) or the 1-median (Brusco, 2017) of each cluster as its representative solution.

The remainder of this section will discuss the features of clustering methods in accordance with the classification given in Fig. 6.

4.3.1. Partitional clustering method

The partitional clustering method, also known as the partitioning method, directly divides data objects into a prespecified number of non-overlapping clusters. Theoretically, the ideal partition according to a specific criterion can be obtained by enumerating all possible combinations (Xu & Wunsch, 2005). However, due to high calculation cost, this strategy may be impractical, especially for a large Pareto front. Thus, heuristic algorithms have been employed to find approximate solutions. Different partitioning methods, albeit based on different criteria, follow similar standard procedures as shown in Algorithm 3. Four partitioning methods found in the previous literature on Pareto pruning are described below.

Algorithm 3. Partitional clustering methods

Input: Pareto optimal set, number of clusters (K), partitioning criterion, termination condition

Output: pruned set (representative solutions) and clusters of solutions

- 1: standardize objective values of solutions in Pareto optimal set
- 2: randomly choose K standardized objective values to identify initial clusters
- 3: **while** termination condition is not satisfied **do**
- 4: (re)arrange each data point to identify new clusters
- 5: update cluster membership and partitioning criterion value
- 6: update termination condition
- 7: **end while**
- 8: (if necessary) identify representative solutions
- 9: convert standardized back to original objective values
- 10: return pruned set and clusters of solutions

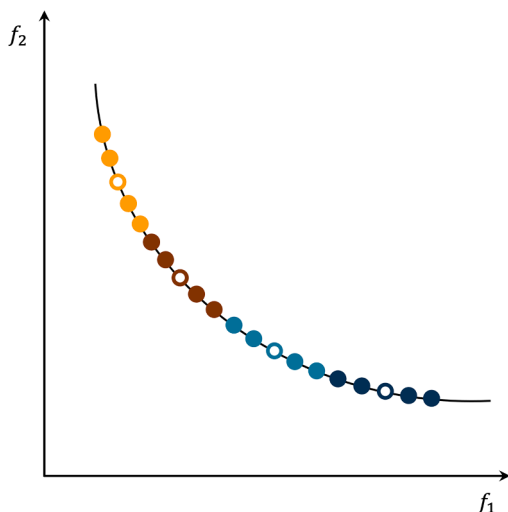


Fig. 5. Pruned set and clusters of Pareto optimal solutions (adapted from Petchrompo et al. (2022)).

- **K-means clustering:** In the K -means method, the partitioning is performed with an objective to minimize the sum of squared deviation (Euclidean distance) of data points to the centroids of their clusters (MacQueen, 1967). The value of each centroid is continuously updated by calculating the mean value of cluster members in each iteration. Since the calculated centroid does not usually correspond to an actual data point, an additional step is required to identify the representative solutions. Due to the efficiency of the algorithm and the interpretability of the results, the K -means method is the most widely-used partitioning method in the Pareto pruning literature (Taboada & Coit, 2007; Taboada et al., 2007; Mahdavian, Sudeng, & Wattanapongsakorn, 2017; Sato, Izui, Yamada, & Nishiwaki, 2019; Lal & Bithin, 2019).
- **K-medoids clustering:** As with the K -means method, the K -medoids method also seeks to minimize the distance between data points and a selected cluster center. The difference between the two is the process of identifying the cluster center. Specifically, the K -medoids method uses an actual data point (medoid), rather than a calculated mean (centroid), as the cluster center (Kaufmann & Rousseeuw, 1987). In each iteration, a non-medoid solution is selected as a new candidate, and other solutions are reorganized into new clusters according to a selected distance measure. The use of actual solutions as cluster centers offers more flexibility because any distance or similarity metrics can be adopted. The Euclidean distance and cosine similarity were employed in the related literature (Ashbolt, Maheepala, & Perera, 2017; Li, Xie, Kim, & El-Basyouny, 2019; Petchrompo et al., 2022).
- **Maximum-split partitioning (MSP):** The MSP method partitions the data by maximizing dissimilarity of the solutions that belong to different clusters. More precisely, the clusters are organized in such a way that the smallest distance between two solutions from different

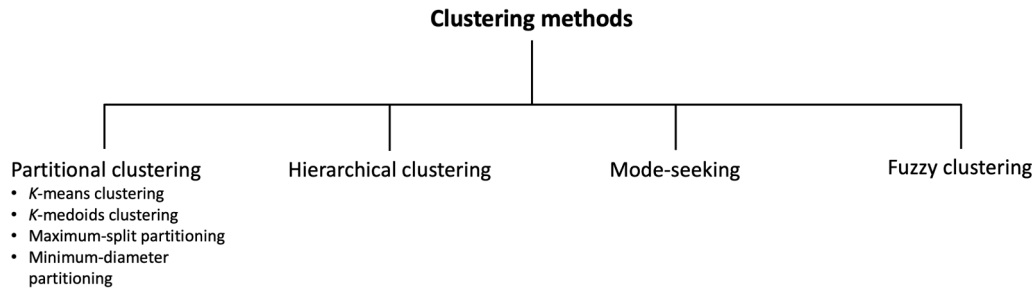


Fig. 6. Classification of clustering methods for Pareto pruning.

clusters is maximized (Hansen, Jaumard, Meyer, Simeone, & Doring, 2003). Since the representative solution is not directly provided by the MSP method, a solution located in the cluster center should be used as the representative solution. More details of the application of the MSP method in Pareto pruning can be found in Brusco (2017).

- **Minimum-diameter partitioning (MDP):** In contrast with MSP, the MDP method aims at maximizing the similarity of the solutions within the same cluster. In other words, the partitioning is performed by minimizing the cluster diameter — the largest distance between two solutions within that cluster (Brusco & Steinley, 2014). The MDP method also requires an additional step for identifying the representative solutions. The application of the MDP method to partitioning Pareto optimal solutions was demonstrated in Brusco (2017).

4.3.2. Hierarchical clustering method

In the hierarchical clustering (HC) method, the data points are grouped by creating a tree-like nested structure. Unlike the partitioning method which produces a single set of clusters, the HC method constructs a hierarchy in which data points are joined together at different levels (Xu & Wunsch, 2005). The outputs of the hierarchical clustering methods are generally depicted in a dendrogram.

In the context of Pareto pruning, the root node of the dendrogram represents the Pareto optimal set with each leaf corresponding to a Pareto optimal solution. The intermediate nodes illustrate how close the objective values of different solutions are to each other. It is apparent that the HC method does not require the DM to specify the number of clusters prior to the resolution process. The final clusters are determined after the clustering process by cutting the dendrogram at a selected level. As with MSP and MDP, the HC method does not directly provide the pruned set, as it does not automatically identify the representative solution of each cluster.

The HC method was introduced to pruning the Pareto front in Morse (1980) and was later used as a benchmark in Zhang, Wu, and Chen (2014). A specific hierarchical clustering method called dynamically growing self-organizing tree (DGSOT) was used to address the all-terminal network reliability problem with three objectives (Aguirre & Taboada, 2011).

4.3.3. Mode-seeking method

The mode-seeking method, also known as the MeanShift method, aims at assigning data points to the clusters iteratively by shifting points towards a selected mode (Comaniciu & Meer, 2002). In the application to Pareto pruning, the mode is the highest density of Pareto optimal solutions in the region. Unlike the partitioning method which adopts a distance metric as the partitioning criterion, the mode-seeking method groups the data points according to the compactness of nearby points. Specifically, in each iteration, these points move gradually towards the area where Pareto optimal solutions are densely populated. At the convergence, the algorithm returns the the center of gravity for each cluster, and the closest solution to such point is chosen as the cluster representative.

The aforementioned process shows that the mode-seeking method does not require any input from the DM, as the algorithm automatically

determines the appropriate number of clusters with respect to the data characteristics. An interesting application of the mode-seeking method in the post-optimality analysis can be found in Weiss (2019). The author demonstrated how this clustering method can identify a small subset of diverse solutions from a large Pareto front in a composite structure design problem.

4.3.4. Fuzzy clustering method

In all the aforementioned clustering methods, each Pareto optimal solution can belong to only one cluster. The fuzzy clustering method, on the contrary, allows a solution to become a member of multiple clusters at the same time. These clusters can be achieved by assigning membership degrees to Pareto optimal solutions. The membership degree ranges between 0 and 1; a higher value indicates more likelihood that a data point will belong to a cluster. In the literature on Pareto pruning, the fuzzy C-means clustering (FCM) method was applied to a MOO problem in structural engineering (Wang & Jia, 2020). The resolution process of the FCM method, as with the K-means method, revolves around the use of cluster centers in grouping solutions. The difference between the two methods is that the membership matrix and the fuzzy criterion function is updated in each iteration. The closest solutions to the cluster centers are finally chosen as the representative solutions. If hard clusters are required, each solution can be assigned to a cluster with the highest membership degree.

An important advantage of the clustering method over the hypervolume-based method lies in its inherent grouping of non-selected solutions. This feature allows the number of solutions in different groups (clusters) to be more evenly distributed. The grouping is crucial in the decision-making process because the non-selected solutions are not immediately removed from the analysis but are generally revisited once the DM specifies the area of his/her interest through choosing a representative solution.

Different clustering methods also possess different advantages and disadvantages. It is evident that the performance of the partitioning method is reliant on the pre-specified number of clusters, which in turn is the number of representative solutions in the pruned set. On the other hand, the hierarchical clustering method gives the DM an option to choose the number of clusters from the resultant dendrogram, while the clustering process in mode-seeking method depends solely on the compactness of nearby solutions. Moreover, in the existing literature, there has yet to be a study that indicates a distinct advantage of the fuzzy clustering methods over the others in the context of Pareto pruning. This aspect should therefore be investigated in a future study.

4.4. Dimensionality reduction method

The dimensionality reduction method is based on the assumption that some of the objectives considered in the MOO problem may be redundant. This means that some solutions in the same region of the Pareto front can be removed without significantly affecting the topology of the Pareto front. By reducing the dimension of the objective space, essential representative solutions from different parts of the Pareto front can be presented to the DM in the preliminary decision-making stage.

Dimensionality reduction techniques remove such redundancies by projecting high-dimensional data onto a low-dimensional space. The transformation is performed in such a way that key properties of the data are retained. In the existing Pareto pruning literature, these techniques were shown to effectively address the problem with many objectives, because the analysis of the Pareto front in more than three dimensions is usually intractable.

Previous researchers demonstrated the applications of the self-organizing map (SOM) and the principal component analysis (PCA) in the post-optimality analysis of their pruning methods. For a system reliability problem, Li et al. (2009) applied SOM to pruning the Pareto front obtained from NSGA-II. The transformation was performed on the basis of similarity using both Euclidean distance and the cosine similarity by continuously updating the weight vectors. This resulted in the projection of the three-dimensional Pareto front onto a two-dimensional space, facilitating the cluster analysis on the Pareto optimal solutions. Sabio, Kostin, Guillen-Gosalbez, and Jimenez (2012) concurrently optimized six environmental metrics of hydrogen infrastructures using the ϵ -constraint method and employed PCA to help interpret the resultant Pareto optimal solutions. This process was shown to effectively eliminate redundant objectives in the post-optimality analysis. This ultimately allowed the DM to be familiarized with a diverse yet manageable set of Pareto optimal solutions.

It is evident that the output produced by the dimensionality reduction method is different from the results obtained from the other three. Specifically, this technique only reduces the number of objectives of the MOO problem but does not directly identify the representative solutions that constitute the pruned set. Thus, a distinct advantage of the dimensionality reduction method is that it can disclose hidden relationships between objectives, thereby allowing the DM to considerably filter out unnecessary solutions in the preliminary analysis. Nonetheless, a drawback of this method is that it may be ineffectual at producing reasonable results if the problem does not involve redundant objectives. More precisely, it could be even more difficult for the DM to comprehend the results in the two dimensional space because their axes, for example the principle components in PCA, are not always interpretable.

5. Efficiency-based pruning method

The primary objective of methods in this class is to identify a small set of solutions that are efficient as compared to other solutions in the Pareto optimal set. Specifically, in the absence of the preference information, it is assumed that the solutions with high marginal rate of return among the conflicting objectives is more attractive to the DM (Bechikh, Ben Said, & Ghédira, 2011; Liu, Li, Zhao, & Liu, 2019). In other words, the key concept of the efficiency-based methods is to find the solutions that yield better trade-offs than other solutions on the Pareto optimal front. The pruning instruction of methods in this class is based on one or more efficiency indicators specified by the DM. Having delved into the relevant literature, we can identify four major groups of efficiency-based pruning methods as shown in Table 4. These methods will be further discussed in the remainder of this section.

5.1. Knee-based pruning method

One widely used strategy in efficiency-based pruning is to identify the solutions located at frontier knees. The frontier knees, also known as the knee regions, are parts of the Pareto front with high trade-off ratios

between objectives (Branke et al., 2004); these are conceptually the most bulging regions on the Pareto front (Lee, Lee, Ahn, & Choi, 2018). In other words, frontier knees can be considered parts of the Pareto front where a small improvement in one objective could dramatically deteriorate at least one other objective. Examples of knee solutions in two-dimensional space are illustrated in the white markers shown in Figs. 7–9.

According to the definitions of knees in the existing literature, the knee-based method generally works on the assumption that there exists a bulge on the Pareto front. This means that the knee-based method is fruitless when applied to a linear Pareto front.

There are two main rationales behind the use of frontier knees in the pruning process. First, since only a minor positive change in one objective could severely worsen other objectives, moving away from a knee solution in any direction is not a logically coherent strategy (Zhang et al., 2015). This is because solutions in the knee regions are inherently more efficient than their neighboring solutions. Second, the findings from previous studies indicated that knee solutions are often located in the middle of the Pareto front, thereby serving as a good compromise among the conflicting objectives (Hosseininassab, Shetab-Boushehri, Hejazi, & Karimi, 2018).

Despite the common goal of finding efficient representative solutions, there are slightly different variations in the way frontier knees are mathematically interpreted. According to the definitions discussed in the previous studies, we can identify three variations of knees: CHIM-based, trade-off-based, and angle-based knees.

5.1.1. CHIM-based knee

In the first variation, Das (1999) defined a frontier knee in a bi-objective minimization problem as the point that is located furthest from the convex hull of individual minima (CHIM) — the boundary line that connects the two ending points of the entire Pareto front — in the direction towards the origin. This variation of frontier knees is referred to as CHIM-based knees in this review. Based on its definition, this pruning method can identify only one single solution, known as the global knee solution. This variation of frontier knees is illustrated in Fig. 7. To incorporate more solutions in the pruned set, later researchers generalized this concept by redefining frontier knees as points on the Pareto front that yield the maximum distance from CHIM in their neighborhood (Hosseininassab et al., 2018).

5.1.2. Trade-off-based knee

In the second variation, the trade-off information provided by the DM is deployed to identify the knee regions. In this paper, we use the

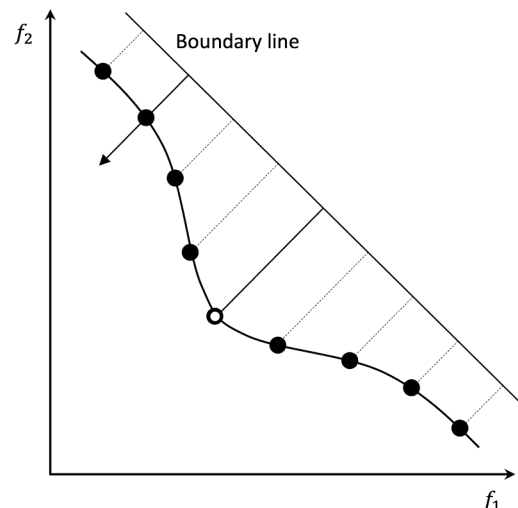


Fig. 7. CHIM-based knee in two-dimensional space (adapted from Das (1999)).

Table 4
Efficiency-based pruning methods.

Methods	Intra-optimality	Post-optimality
Knee-based	✓	
Data envelopment analysis (DEA)		✓
Hypervolume-based		✓

term trade-off-based knees to describe this variation of knees. This is based on the rationale that a point is considered a knee point if there is no other point that surpasses a desired threshold (Deb & Gupta, 2011). Specifically, for a point in a bi-objective problem (f_1 and f_2), the DM is only interested in moving away from this point only with two trade-off conditions: (i) at least α unit sacrifice in f_2 for one unit improvement in f_1 and (ii) at least β unit sacrifice in f_1 for one unit improvement in f_2 . This means that the identification of knees is based on the DM-specified parameters α and β . Therefore, the points on the Pareto front that satisfy such conditions are called the (α, β) -knee points, as shown in Fig. 8. This variation of knees also lays the foundations for the concept of cone-dominance, as the points in the (α, β) -knee region are said to cone-dominate other points on the Pareto front (Ramirez-Atencia, Mostaghim, & Camacho, 2017).

5.1.3. Angle-based knee

As for the third variation, a frontier knee can be determined by the angle between a point on the Pareto front and at least two neighbors. We refer this type of frontier knees to as the angle-based knees. According to the methods employed in the related literature, three main concepts of the angle-based knees can be identified:

- **Reflex angle:** Early adopters of angle-based knees (Branke et al., 2004) proposed constructing two lines between a selected point and two adjacent points on the Pareto front. The angle between those lines is then measured. As shown in Fig. 9, an individual with a large angle measure (γ) is considered a frontier knee. The term reflex angle was later used to explain this concept in a more recent article (Deb & Gupta, 2011). Branke et al. (2004) also suggested another variant of the reflex angle in which four nearest neighbors (two from each direction) are used to measure the reflex angle. In this case, four angles are determined by the combination of lines that connect any point to the left and that to the right at a time. The largest angle among these four is chosen as the representative reflex angle.
- **Bend angle:** The application of the bend angle in identifying knee regions was introduced by Deb and Gupta (2011). Instead of using the closest neighbors to a selected point as in the calculation of the traditional reflex angle, the bend angle is based on the normalized objective values of any two points that are located on the left and right of a selected point on the Pareto front. This means that the two reference points need not be adjacent to the point of interest but depend on the discretion of the DM. In addition, the authors also proposed comparing the calculated angle with a predefined threshold. More specifically, a point on the Pareto front is considered

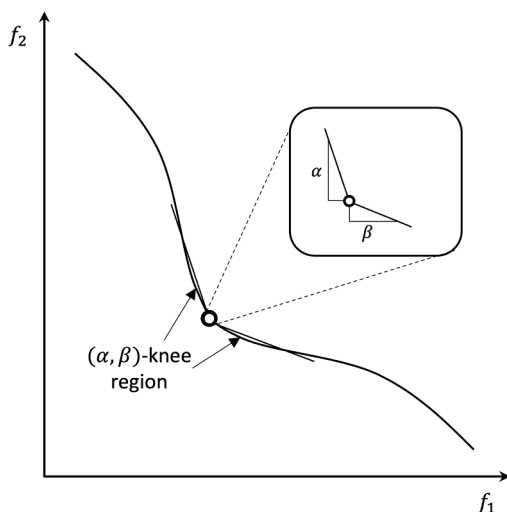


Fig. 8. Trade-off-based knee in two-dimensional space (adapted from Deb and Gupta (2011)).

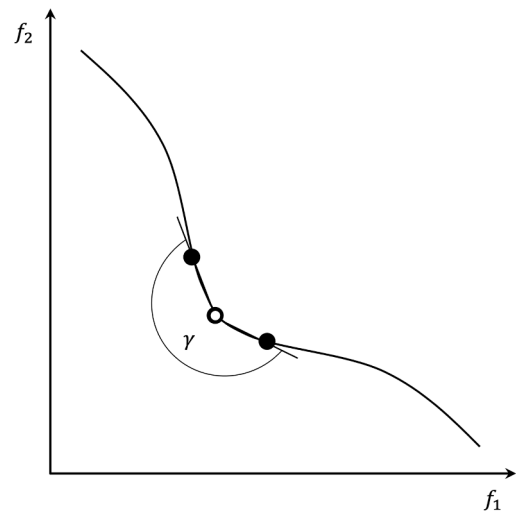


Fig. 9. Angle-based knee in two-dimensional space (adapted from Branke et al. (2004)).

a bend-angle-based knee point only if its bend angle is greater than the DM-supplied threshold. Despite its flexibility in the selection of reference points, the application of the bend angle also possesses a drawback in that the existence of a knee point is not guaranteed, especially if a poor choice of the reference threshold is made.

- **Extended angle:** The concept of extended angle was first proposed by Sudeng and Wattanapongsakorn (2013) who developed the adaptive angle-based pruning algorithm (ADA). The extended angle dominance is based on the comparison between a calculated reflex angle and a reference angle called the delta (δ) angle. Unlike the reference threshold applied in the bend angle, the delta angle is not supplied by the DM but autonomously adjusted using a crowding estimator method. This concept has laid the foundations for later knee-based pruning methods such as angle-based with specific bias parameter pruning algorithm (ASA) (Sudeng & Wattanapongsakorn, 2015) and ranked ADA (RADA) (Choachaicharoenkul & Wattanapongsakorn, 2020). ASA enhances the flexibility of ADA through the introduction of an additional pruning instruction which allows the DM to adjust the number of solutions in the pruned set. On the other hand, RADA does not filter out but imposes the penalty on the angle-dominated solutions. The total angle penalty score is then used to rank the solutions and finalize the pruned set.

It is apparent that the pruning instruction in the knee-based methods is generally applied during the optimization process. More precisely, these studies embed the pruning instruction directly into their optimization algorithms. The objective of such intra-optimality methods is to guide the solutions towards knee regions, thereby reducing the computational cost in their search process. Some researchers developed their own evolutionary algorithms to locate frontier knees. Rachmawati and Srinivasan (2006) coupled a bias selection criterion with weighted-sum functions to find knee regions in the objective space. Latterly, Zhang et al. (2015) developed a knee-point-driven evolutionary algorithm by using knee points in the parent selection step of their algorithm. In a recent study, Zhang, Shen, He, and Yen (2021) devised the knee-based multimodal multi-objective evolutionary algorithm, which aimed to search for a complete set of global knee solutions. On the other hand, some researchers customized existing MOEAs by including the knee-based pruning instruction in their search algorithm. Sudeng and Wattanapongsakorn (2016) integrated the ASA concept with NSGA-II and called their customized algorithm the k-ASA-NSGA-II algorithm. In their algorithm, the ASA is applied to eliminate inefficient solution in each iteration, allowing the solutions to converge to the frontier knees. Other studies that tailored existing MOEAs to search towards knee regions

include Branke et al. (2004), Bechikh et al. (2011), Ramirez-Atencia et al. (2017), Li, Wang, Zhang, Ming, and Li (2020).

The knee-based method is the most acknowledged technique in this class mainly because its outputs are easily interpretable. It is readily understandable to the DM why the solutions in the bulging region of the Pareto front are included in the pruned Pareto set. Despite the variations of frontier knees in the literature, different knee-based methods generally provide similar results. A major drawback of the knee-based approach lies in its inefficacy when applied to high-dimensional problems. From the discussion above, we can see that some knee-based indicators are limited only to bi-objective problems, while some are very computationally costly as more objectives are included in the resolution process.

5.2. Data envelopment analysis method

To locate efficient solutions in the Pareto optimal set, previous researchers also applied the data envelopment analysis (DEA) in their post-optimality stage. More specifically, the DEA-based pruning method is generally used to reduce the number of solutions after the entire Pareto front is obtained from the *a posteriori* method (Khalili-Damghani, Abtahi, & Taviana, 2014; Li, Mobin, & Keyser, 2016b). DEA is a nonparametric method based on linear programming that was originally used to evaluate the productive efficiency of decision-making units (DMUs) Charnes, Cooper, and Rhodes (1978). The primary objective of DEA is to bolster the decision-making process in which each decision is associated with multiple inputs and outputs. Despite its primary application in the production theory, DEA was later employed to benchmark the performance of service operations in various fields (Cook & Seiford, 2009; Liu, Lu, Lu, & Lin, 2013).

DMU is defined as a unit whose performance, the relative efficiency (RE), is determined by the ratio between its weighted sum of outputs and weighted sum of inputs. In benchmarking, multiple DMUs are compared and ranked by their RE values. Most efficient DMUs (DMS with high rankings) then constitute the DEA efficient frontier.

DEA was first applied to Pareto pruning by Li et al. (2009) to address a system reliability optimization problem. For pruning purposes, each solution on the Pareto front is treated as a DMU. Multiple objectives are categorized into inputs and outputs for the DMU. Cost-type objectives such as monetary cost and service disruption time are classified as inputs, whereas benefit-type objectives such as system reliability and production rate are classified as outputs. To justify the use of RE, these DMUs are generally assumed to be homogeneously comparable. Consider a Pareto optimal set of p solutions, each of which is associated with m inputs x_{ik} ($i = 1, 2, \dots, m$) and n outputs y_{jk} ($j = 1, 2, \dots, n$). The relative efficiency of the k^{th} DMU, RE_k ($k = 1, 2, \dots, p$), is given by:

$$RE_k = \frac{\sum_{j=1}^n u_j y_{jk}}{\sum_{i=1}^m v_i x_{ik}} \quad (3)$$

where $u_j \geq 0$ and $v_i \geq 0$ are the weights for the output j and input i respectively.

To achieve the maximum RE value for each DMU, a specific set of input and output weights must be identified. It is apparent that the maximization of this ratio depends on the relative magnitude of the weighted sum of outputs (the numerator) and the weighted sum of inputs (the denominator). To make the problem solvable by a linear programming algorithm, the denominator can be normalized by setting its value equal to a constant ($= 1$) (Li, Hu, Zhou, Li, & Kang, 2018). The optimization problem is then transformed into the maximization of the numerator, and the maximum RE of solution k can finally be obtained by solving the following problem:

$$\begin{aligned} & \text{maximize } RE_k = \sum_{j=1}^n u_j y_{jk} \\ & \text{subject to :} \\ & \quad \sum_{i=1}^m v_i x_{ik} = 1 \\ & \quad \sum_{j=1}^n u_j y_{jk} - \sum_{i=1}^m v_i x_{ik} \leq 0, \quad k = 1, 2, \dots, p \\ & \quad u_j, v_i \geq \varepsilon, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n \end{aligned} \quad (4)$$

where ε is a non-Archimedean element smaller than any positive real number.

Since the objective function and constraints are different for each DMU, a linear program for each Pareto optimal solution has to be solved separately. It is apparent that the linear program in Eq. (4) aims at maximizing the RE value. Hence, previous researchers suggested that solutions with high RE values are included in the pruned set (Li et al., 2009). We will explicate how the value of RE is interpreted in Section 7, where performance indicators for the pruned set are discussed. Other interesting applications of the DEA-based pruning method can be found in Karamouz and Nazif (2013), Ustun and Anagun (2015).

It is apparent that the DEA-based method can effectively deal with high-dimensional problems as it employs the relative efficiency indicator in the pruning process. This indicator is based on the ratio between weighted sum values of objectives, and the representative solutions are determined by linear programming. Hence, the DEA-based method can deal with any number of objectives. However, the pruning process of such method can be very time-consuming if the Pareto optimal set contains a large number of solutions. This is because the number of Pareto optimal solutions contributes directly to the number of constraints in the linear programming model.

5.3. Hypervolume-based method

As previously discussed in Section 4.1, the hypervolume metric can be used to measure both the diversity and the convergence of Pareto optimal solutions according to the reference point prerequisite. Hence, this metric was also employed in the context of efficiency-based pruning in some previous literature. Although the hypervolume-based method is applied in the post-optimality stage of both diversity-based and efficiency-based pruning methods, the major difference between the two is the number of solutions used in the hypervolume calculation. In the diversity-based method, multiple representative solutions are used in the calculation of the hypervolume. On the other hand, in the efficiency-based method, the hypervolume of a single solution is considered. Specifically, the method in this class follows the rationale that the solution with the largest hypervolume is the one that is located furthest to the reference point, thereby yielding the highest trade-off ratio.

The largest-hypervolume method was applied to supporting decision-making on identifying key players in social network (de la Fuente et al., 2018) and summarizing multi-document text (Sanchez-Gomez, Vega-Rodriguez, & Perez, 2019). In both applications, the hypervolume-based method was demonstrated to effectively reduce the Pareto front to one representative solution.

An observation from the literature discussed in this section indicates that the final results obtained from the hypervolume-based method are consistent with other efficiency-based methods such as the knee-based method. However, the hypervolume-based method can be very computationally expensive in a problem with many objectives (Zitzler, Thiele, Laumanns, Fonseca, & Da Fonseca, 2003a). Furthermore, the pruning process can be very cumbersome because, in its application to finding efficient solutions, the hypervolume of every single solution has to be determined separately.

6. Problem-specific pruning method

In the three classes of pruning methods discussed in Sections 3–5, the pruning instructions are based on the measures that are universally applicable to any MOO problem. These instructions can be derived either from the preference of the DM, who is always pertinent to the decision-making process, or from the topology of the Pareto front (e.g. well-distributed solutions in diversity-based pruning or knee regions in efficiency-based pruning). In the problem-specific pruning method, on the contrary, the pruning instructions are tailored to a particular problem. Therefore, methods in this class are based on the assumption that there exists a specific criterion that is independent of the objective functions.

A fine example of a method in this class can be found in [Pehrsson, Ng, and Stockton \(2013\)](#) in the context of production system management. To reduce the number of Pareto optimal solutions obtained from their simulation-based MOO model, the authors developed an independent indicator called the pay-off filter, which is the length of time required to recover the initial cost of an investment. This indicator, independent of the objectives considered in the optimization model, allows the DM to shortlist promising solutions for the pruned set. Another interesting example of a problem-specific pruning method found in the literature is the use of customized subjective criteria as the pruning instruction. [Ahmed, Deb, and Jindal \(2013\)](#) developed a novel approach to bolstering the decision-making process for cricket team selection. To prune the Pareto optimal solutions which maximized different types of team performance, the authors incorporated a subjective factor such as the selection of star players into the post-optimality analysis.

In a stochastic problem in which the exact values of Pareto optimal solutions are uncertain, previous researchers also derived the pruning instruction from the DM's attitude towards risk. [Jornada and Leon \(2016\)](#) employed a random vector to represent the multiplicative perturbation to develop the robustness projection set and the robustness space. These concepts allowed the DM to break ties among the Pareto optimal solutions by identifying the pruned set that is less sensitive to uncertainties. [Felfel, Ayadi, and Masmoudi \(2016\)](#) embedded the concept of the downside risk into the objective functions of their stochastic MOO model to yield a pruned set of robust solutions. Later on, [Cao et al. \(2020\)](#) also incorporated uncertainties into the decision-making process to deal with their reliability allocation problem. In their study, the DM is required to identify the best solution from a large Pareto optimal set which represents the trade-off between the system reliability and the cost of a series-parallel system. Their problem is formulated under the assumption that the reliability and the cost of each component are uncertain. The game-theory-based selection method was applied to help the DM with this task. The objectives and the Pareto optimal set obtained from NSGA-II were considered to be the players and the strategy set respectively. The utility function was derived from the normalized system entropy, which provides a measure of randomness. By finding the Nash equilibrium of the game, the DM can arrive at the Pareto optimal solution with a relatively high entropy and low associated variance.

The problem-specific pruning method allows the DM to apply a filter to greatly reduce the number of Pareto optimal solutions in a meaningful way. Nonetheless, not every problem can be addressed by this method because it is not always possible to identify a sensible criterion that is independent of the objectives in the optimization process.

7. Performance indicators for the pruned set

There are two main reasons behind the need for performance indicators for the pruned set. First, there is a need for a solid evidence to support the DM in the selection of a conducive pruning method for a particular problem. This means that a proper indicator should be deployed to compare different pruning methods. Second, predefined inputs are required for certain pruning methods (e.g. the number of

representative solutions in the pruned set for a partitioning method). The DM is therefore in need of a reliable indication to help determine appropriate input values.

Previous researchers concurred that there has yet to be any indicator that can signify the superiority of a pruned set over another in every aspect ([Branke, 2008](#); [Sudeng & Wattanapongsakorn, 2015](#); [Petchrompo et al., 2022](#)). It was also evident in Sections 3–6 that different pruning methods were developed for different purposes, thereby demonstrating the need for different indicators for the pruned set. In this review, we elaborate on the indicators compiled in [Petchrompo \(2021\)](#). Performance indicators for the pruned set classified by the pruning purpose are exhibited in [Table 5](#). It is important to note that some indicators adopted in the previous literature are similar to those for the approximate Pareto optimal set ([Zitzler, Thiele, Laumanns, Fonseca, & da Fonseca, 2003b](#)), but we will discuss only their application to the pruned set and how they can be interpreted in the remainder of this section.

- **Utility-based measure:** The utility-based measure (UBM) for the pruned set was proposed by [Branke et al. \(2004\)](#). The indicator is based on the concept of the expected marginal utility. In the context of Pareto pruning, the utility is defined as the DM's value of

Table 5
Performance indicators for the pruned set.

Indicator	Pruning purpose			Calculation
	Preference	Diversity	Efficiency	
Utility-based measure (Branke et al., 2004)	✓			Expected marginal utility elicited from a DM
Relative closeness value (Yadollahi et al., 2015)	✓			Distance between ideal and representative solutions
Silhouette index (Taboada & Coit, 2007)		✓		Average Euclidean distance between intra-cluster and inter-cluster
Ratio indicator (Brusco, 2017)		✓		Improvement in a distance measure when the number of representative solutions increases
Relative Euclidean Index (Petchrompo et al., 2022)		✓		Average relative Euclidean distance between individual and representative solutions
Relative cosine Index (Petchrompo et al., 2022)		✓		Average relative cosine similarity between individual and representative solutions
Hypervolume (Sanchez-Gomez et al., 2019)		✓	✓	Volume of the objective space enclosed by reference points and representative solutions
Angle-based indicator (Branke et al., 2004)			✓	Angle between an individual solution and neighboring solutions in the objective space
Relative efficiency (Li et al., 2009)			✓	Ratio between weighted sum of benefit-type and cost-type objective values
Generational distance (Zhang et al., 2021)			✓	Average Euclidean distance between representative and closest true Pareto optimal solutions

satisfaction on a particular solution; hence, the marginal utility is defined as the additional cost that the DM would accept if he/she had to settle for the second preferred solution. The expected marginal utility for a solution can be determined by integrating over all possible utility functions elicited from the DM in a bi-objective problem. For a case of more than two objectives, this value can be approximated by sampling. A representative solution with a high UBM value, that is the individual with large overall marginal utility, is preferred by the DM.

- **Relative closeness value:** The relative closeness value (RCV) was applied in [Yadollahi et al. \(2015\)](#). The calculation of RCV is based on the DM's importance weights for different objectives and ideal points. In RCV, the normalized weights are elicited from the DM using the AHP method. The weighted normalized decision matrix is then used to determine the positive and negative ideal solutions. RCV for each representative solution is finally derived from the distance between each representative and the ideal solutions. The range of RCV is from 0 to 1. RCV close to 1 signifies that the representative solution is preferred by the DM, whereas RCV close to 0 indicates that the representative solution is undesirable.
- **Silhouette index:** The silhouette index, also known as the global silhouette value, was first applied to Pareto pruning in [Taboada and Coit \(2007\)](#). To evaluate the quality of the clustered solutions, the silhouette index measures the similarity of a solution to other solutions within the same cluster compared to those from a neighboring cluster. The value of the silhouette index is derived from the average cluster silhouette, which indicates heterogeneity and isolation properties of each cluster. Each cluster silhouette is determined by averaging all silhouette widths within the corresponding cluster. The silhouette width of an individual solution is calculated by normalizing the average Euclidean distance from that solution to all the observations within the same cluster against the average Euclidean distance from that solution to all the observations that belong to the nearest cluster. The value of silhouette width ranges from -1 to 1 . A value close to -1 signifies that the solution is not assigned to an appropriate cluster, while a value close to 1 indicates that the solution is distinct from the solutions from the adjacent cluster. This leads to the conclusion that a higher silhouette index value is better, as it demonstrates that the cluster assignment is effective.
- **Ratio indicator:** To compare the performance of pruned sets with different numbers of representative solutions, the ratio indicator was suggested by [Brusco \(2017\)](#). The ratio indicator determines the improvement in a measure for partitioning. For instance, the average

Euclidean distance between the representative solutions and all other solutions in their cluster is used as a measure for the K -means method. The ratio indicator calculates the ratio between an improvement in the selected measure when moving from $K-1$ to K clusters with that when moving from K to $K+1$ clusters. A higher value of ratio indicator suggests that the improvement is more satisfactory. The developer of this indicator therefore recommended to choose the number of representative solutions that yield the maximum ratio indicator value.

- **Relative Euclidean index:** The relative Euclidean index (REI) was developed by [Petchrompo et al. \(2022\)](#). The calculation of REI is based on the Euclidean distance metric. The essence of REI lies in two major aspects. First, REI measures the average Euclidean distance between the representative solutions and other solutions within the cluster. Second, the calculated distance is normalized against two benchmark scenarios shown in [Fig. 10](#) where black and white markers symbolize the Pareto optimal and the representative solutions respectively. The worst-case scenario is the case in which one central representative solution represents the entire Pareto front, while the best-case scenario is the case in which all solutions on the Pareto front are captured in the pruned set. The range of REI is from 0 to ∞ . A low REI value (close to 0) indicates that the pruned set can strongly represent the entire Pareto front. In contrast, a REI value greater than 1 shows that the diversity of the pruned set is poor as it is outperformed by a single representative solution.
- **Relative cosine index:** The rationale behind the relative cosine index (RCI) is similar to that behind REI. Specifically, RCI also indicates whether the pruned set strongly represents other non-selected solutions ([Petchrompo et al., 2022](#)). As with REI, RCI is also normalized against two benchmark scenarios shown in [Fig. 10](#). The difference between the two indicators lies in the use of the cosine similarity in RCI instead of the Euclidean distance in REI. Due to the adoption of this similarity metric, the interpretation of RCI is in the opposite direction to that of REI. The RCI value ranges from $-\infty$ to 1 . A RCI value close to 1 suggests that the representative solutions are similar to other solutions within the same cluster, thereby signifying the diversity of solutions in the pruned set. In contrast, a negative RCI value indicates that the chosen pruned set poorly represents the Pareto optimal solutions, as it demonstrates that the diversity of the pruned set is even worse than that of a single central solution.
- **Hypervolume:** As previously discussed in Section 4, the hypervolume metric is originally used to measure the performance of the Pareto optimal set. In its application to evaluating the pruned set, the

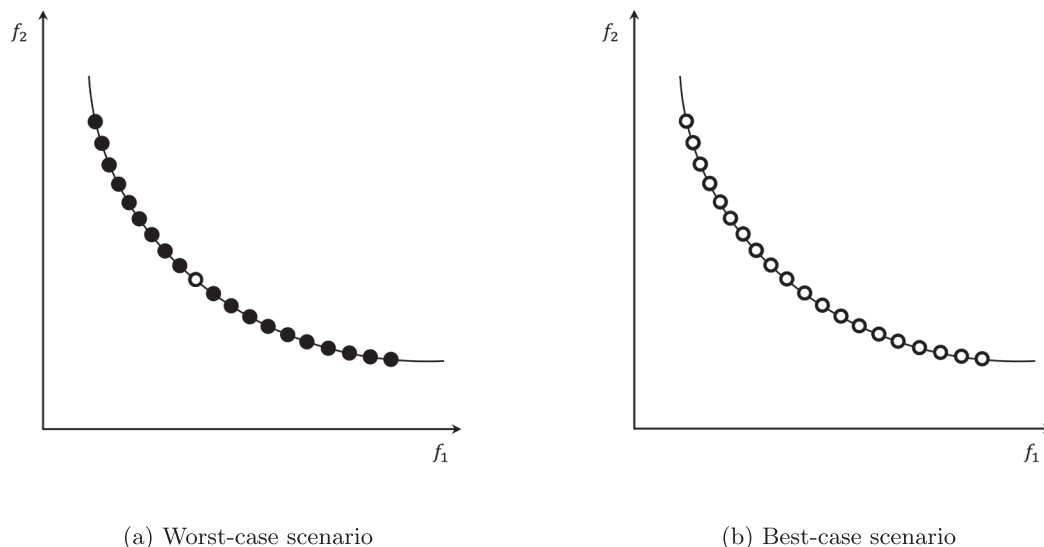


Fig. 10. Benchmark scenarios for relative diversity indicators (adapted from [Petchrompo et al. \(2022\)](#)).

hypervolume metric is defined as the objective space enclosed by a reference point and the pruned Pareto front. Previous researchers demonstrated the use of hypervolume to measure both the diversity (Petchrompo et al., 2022) and the efficiency (Sanchez-Gomez et al., 2019) of solutions in the pruned set. A high hypervolume value indicates that the pruned Pareto front is located distantly from the reference point, thereby suggesting that the representative solutions are diverse and/or efficient.

- **Angle-based indicator:** Developers of some knee-based pruning methods employed the angles mentioned in Section 5.1 to evaluate the pruned set (Branke et al., 2004; Deb & Gupta, 2011; Sudeng & Wattanapongsakorn, 2013). Most angle-based indicators are developed upon the reflex angle — the angle between a selected solution and its neighbors on the left and right. The interpretation of these indicators is apparently straightforward. A large angle value indicates that the trade-off ratio at a specific representative solution is high when compared to specific neighbors, that is, it suggests that the solution of interest is more efficient than other solutions on the Pareto front.
- **Relative efficiency:** The relative efficiency (RE) is the ratio between the weighted sum of outputs and the weighted sum of inputs as explained in Eq. (3). For each solution, the outputs and the inputs are referred to as the benefit-type and cost-type objective values respectively. RE was employed by previous researchers to measure the solutions using the DEA-based method (Li et al., 2009; Khalili-Damghani et al., 2014). It can be seen from Eq. (4) provided in Section 5.2 that RE ranges between 0 and 1. A low RE value (close to 0) suggests that low outputs are produced by large inputs, signifying the inefficiency of a solution. In contrast, a superior RE value (close to 1) indicates that high output values can be achieved efficiently by nominal inputs. In other words, it suggests that the solution yields high trade-offs between conflicting objectives when compared to other solutions.
- **Generational distance:** In the MOEA community, generational distance (GD) was introduced for measuring the convergence of Pareto optimal solutions in the objective space (Zitzler et al., 2003b). In its original application, GD is used to determine the average distance between the non-dominated vectors generated by a selected MOEA and the true Pareto front. In the context of the pruned set evaluation, GD finds the average of the distances between the solutions on the pruned Pareto front and their closest solution on the true Pareto front. This indicator was usually adopted to evaluate the performance of knee-based methods in the existing literature (Sudeng & Wattanapongsakorn, 2015; Zhang et al., 2021). As a distance-based metric, a low GD value is preferred; a GD value close to 0 means that solutions in the pruned set is less distant to the true Pareto front, thereby signifying a high trade-off ratio between the objectives.

8. Discussion

After the discussion of past research questions, underlying assumptions, methods and findings of individual methods in each of the pruning classes, this section gives an analysis of the traditional MOO and pruning methods when integrated together. This section is comprised of two subsections. First, we compare the features of MOO methods across different classes. The findings from this comparative analysis then enable us to put forward suggestions on the selection of conducive methods for different decision-making conditions. Second, we identify gaps in the current research on MOO and pruning methods as well as offering potential avenues for future work.

8.1. Comparison of multi-objective optimization classes

In this section, we identify the advantages and disadvantages of the *a priori*, *a posteriori*, interactive, and pruning methods. The emphasis is

placed on the comparison between the pruning method and other classes. The summary of the comparison between MOO methods is given in Table 6. As previously discussed in Section 2.4, the pruning methods can be divided into three groups: pre-optimality, intra-optimality, and post-optimality. We found that methods in each group generally share similar advantages and disadvantages; therefore, the discussion in this section will be held in the order of these groups. Specific features of each pruning method class (preference-based, diversity-based, efficiency-based, problem-specific) are also discussed.

8.1.1. A priori method

The major advantage of the *a priori* method lies in its compatibility

Table 6
Comparison of multi-objective optimization methods.

Method	Decision-making condition	Advantages	Disadvantages
<i>a priori</i>	The DM can provide precise information for consolidating multiple objectives prior to the optimization process.	• Simple formulation and implementation • Low computational cost • Resulting in single optimal solution • Compatibility with high-dimensional problem	• Results sensitive to changes in inputs • Inaccurate results without complete knowledge on the problem domain
<i>a posteriori</i>	The DM has insufficient prior knowledge about the solution space but can choose one final solution directly from the Pareto front.	• No prior information required from DM • DM provided with complete information on Pareto optimal solutions	• High computational cost • Difficulty in finalizing solution in high-dimensional problem
Interactive	Preference of the DM changes when exposed to different Pareto optimal solutions.	• DM allowed to adjust parameters during resolution process	• Time consuming with poor initial solutions • Possibility of premature termination of resolution process
Preference-based pruning	(i) Prior preference information can be partially extracted from the DM or (ii) MCDA/SRD can be applied after the optimization process.	• Representative solutions matching DM's preference	• Inaccurate results if objectives cannot be prioritized • Incompatibility with MCDA in case of large Pareto optimal set
Diversity-based pruning	The DM wants to be acquainted with the knowledge on the Pareto front with many solutions.	• No prior information required from DM • Well-distributed representative solutions along Pareto front	• High computational cost • Possibility of obtaining large pruned set
Efficiency-based pruning	The DM prefers efficient solutions and can specify a solution efficiency indicator.	• No prior information required from DM • Representative solutions yielding satisfactory trade-offs	• High computational cost • Possibility of obtaining solutions in extreme regions • Some methods incompatible with high-dimensional problem
Problem-specific pruning	The DM can identify an independent secondary criterion to prune the Pareto front.	• No prior information required from DM • Representative solutions meeting specified criterion	• Incompatibility with most problems

with high-dimensional problems. Since the multi-objective problem is rearranged in a single-objective optimization setting, there is no limitation on the number of objectives that can be addressed in this mode. This trait also offers another distinct advantage in that the model can be solved by efficient deterministic algorithms. In other words, an optimal solution is mostly guaranteed with low computational cost. Moreover, the fact that the *a priori* method results in a single optimal solution saves the DM from carrying out additional work in the post-optimality stage.

However, the *a priori* method is not without its drawbacks. Since only one objective function is used to represent multiple conflicting objectives, the *a priori* method could be ineffectual for a problem with highly sensitive parameters. This is because only small perturbations in weights could lead to a completely different solution. Hence, we can put forward the recommendation to use the *a priori* method only if the DM can perfectly articulate either subjective or objective information to provide inputs for the weighting and conversion processes.

8.1.2. A posteriori method

To transcend the weaknesses of the *a priori* method, the *a posteriori* method can generate Pareto optimal solutions without converting multiple objectives into a single objective at an early stage of the resolution process. Thus, the *a posteriori* method is apparently advantageous for the DM who neither possesses prior knowledge about the potential solutions nor has any preference towards the objectives. Since the *a posteriori* method delivers a complete set of Pareto optimal solutions, the DM can be acquainted with the comprehensive solution space before making final decisions.

Nonetheless, a major difficulty in applying *a posteriori* methods lies in its resolution process. On one hand, it is apparent that the use of scalarization methods in the *a posteriori* mode is ineffective in finding the solutions in the non-convex parts of the Pareto front (Yang, 2014; Kaim et al., 2018). On the other hand, the application of MOEAs generally incurs high computational cost and may not produce fruitful results in a problem with more than four objectives (Konak et al., 2006; von Lücken, Barán, & Brizuela, 2014). Moreover, the generation of a large Pareto optimal set also causes a problem for the DM in the selection of the final solution, especially in a high-dimensional problem where the visualization of the Pareto front is not effective. Therefore, we suggest that the DM adopts the *a posteriori* method only for a problem with no more than three objectives.

8.1.3. Interactive method

In the interactive method, the DM is requested to continuously interact with the algorithm in order to search for satisfactory solutions. This feature allows the interactive method to overcome the limitations of the aforementioned methods. Unlike the *a priori* method, the interactive method requests that he/she keeps updating the preference information during the optimization process. This means that the interactive method is practical for the case in which the DM cannot provide precise inputs to the model. Moreover, the fact that the DM interactively involves in the optimization process allows the DM to manage the amount of information he/she receives at a time. This means that the interactive method resolves the difficulty in comprehending a large Pareto optimal set in the *a posteriori* method.

Despite its advantages, the interactive method also has two major drawbacks. First, since the DM's preference is continuously adjusted during the optimization process, it can be very time-consuming for the algorithm to converge, especially with poor initial solutions. Second, the interactive method may not guarantee to arrive at the DM's most preferred solution; this is because the search process could terminate prematurely if the DM finds a solution that is satisfactory to some extent. Thus, the DM is suggested to take into consideration the length of time it may take to reach the definitive solution and to be careful with the choice of termination criteria when adopting the interactive method.

8.1.4. Pruning method

In the pre-optimality method, the DM is asked to provide the preference information prior to the optimization process, as with the *a priori* method. However, the fundamental difference between the two is that the pruning method does not require the DM to determine precise numerical weights for different objectives. For instance, by using partial preference information, the preference-based pruning method does not culminate in a single optimal solution but result in a focused group of solutions in the DM's preferred regions. A major advantage of the pruning method in the pre-optimality mode over the *a priori* method is that it can also be applied in the case in which the DM can provide only partial information. Moreover, applying the pruning instruction prior to the optimization process can also help reduce the computational cost as compared to the *a posteriori* method, because the entire Pareto optimal set needs not be produced. This advantage is also apparent in the intra-optimality method, which commonly customizes a MOEA to search towards the instructed region of the Pareto front. Therefore, we can put forward a suggestion to use the pruning method in the pre-optimality and intra-optimality modes when the required information or the pruning instruction is available in advance of the optimization process.

The post-optimality method, on the other hand, usually incurs high computational cost. This is because the entire Pareto front has to be determined before a pruning instruction is employed to choose promising solutions therefrom. The post-optimality method offers a distinct advantage over the *a priori* method in that they do not require any prerequisite from the DM. Furthermore, by providing the DM with the representative solutions, these methods can effectively resolve the difficulty in finalizing the solutions from the Pareto front — the key issue found in the *a posteriori* method. Nonetheless, methods in different classes are found to have slightly different advantages and disadvantages.

The preference-based method allows the DM to systematically identify a shortlist of promising solutions using either MCDA or SRD methods. However, the preference-based pruning method is practical only in the case of small or medium-sized Pareto optimal set because the DM is still required to evaluate and compare the Pareto optimal solutions. We can see that the major limitation of the preference-based pruning method, albeit applied in the post-optimality stage, is still the dependence on the input provided by the DM.

The resolution process of the diversity-based method, on the other hand, is based solely on the topology of solutions in the normalized objective space. Hence, it offers a major advantage over the *a posteriori* method in that the DM can be familiarized with the entire Pareto optimal solutions merely through a small yet well-diversified pruned set. Nevertheless, the diversity-based method also possesses some major disadvantages. To ensure the diversity of the solutions in the pruned set, especially in a high-dimensional problem, the number of representative solutions suggested by a chosen indicator can be very large. In this case, the DM is suggested to choose the largest number of solutions that he/she feels manageable. After choosing a preferred solution in the primary representative set, the DM still has an option to further drill down to finalize the solution. In sum, we suggest that the DM adopts the diversity-based pruning method when he/she cannot articulate inputs prior to the optimization process and would like to learn more about different parts of the Pareto front before making the decision.

In the efficiency-based method, the DM is only asked to specify a pruning instruction based on an efficiency indicator. In case the DM does not want to learn about the Pareto front, this method can be an attractive alternative, as it results in a set of representative solutions that yield satisfactory trade-offs between conflicting objectives. However, the efficiency-based method also risks the possibility of identifying solutions in extreme regions. Since the pruning process is based solely on a chosen efficiency indicator, it is possible that the representative solutions with high trade-offs are located in an undesirable region of the Pareto front. Moreover, it is apparent that some efficiency indicators such as angle-based knees are incompatible with the problem with many objectives.

Thus, we recommend that the efficiency method is employed only if the DM wants to strictly exclude the preference information from the decision-making process and is willing to accept solutions from any region of the Pareto front.

The problem-specific pruning method allows the DM to apply a filter to reduce the number of Pareto optimal solutions in the post-optimality stage. Nonetheless, not every problem can be addressed by this method, as a sensible criterion cannot always be identified. Hence, we suggest that this method is only applied if the DM can specify a meaningful second criterion such as solution uncertainties to help prune the Pareto optimal set.

8.2. Recommendations for future research

In this section, we identify research gaps found in the current studies and put forward recommendations for future research.

- Pruning methods for addressing many-objective problems:** Dealing with more than four objectives has long been a challenging problem in the MOO research community, as most MOEAs in the current literature scale poorly with the number of objectives (López Jaimes & Coello Coello, 2015; Li, Li, Tang, & Yao, 2015). Concurrent optimization of many objectives also leads to a large number of solutions required to represent a Pareto front. The findings of this review paper also revealed that many pruning methods are also ineffective in addressing many-objective problems due to either a large number of dimensions or massive size of the Pareto optimal set. This gap therefore provides researchers an opportunity to develop a pruning method that can reduce the number of objectives at an early stage of the optimization process. Hence, we recommend that future studies consider using a dimensionality reduction technique prior to or during the optimization process to develop a novel pre- or intra-optimality pruning method. To do so, researchers are suggested to investigate the application of such techniques as redundancy- or diversity-based supervised filter models (Cai, Luo, Wang, & Yang, 2018).
- Pruning methods based on other diversity optimization:** As discussed in Section 4, hypervolume was employed as an indicator for the pruning instruction in some diversity-based pruning methods. The hypervolume-based method was demonstrated to produce a pruned set of well-distributed and converged solutions in a low computational cost when compared to other post-optimality methods. In the MOO research community, there exists other diversity optimization methods that allow the user to control the number of solutions that will be kept in the Pareto optimal set. However, there has yet to be any research that explicitly applies diversity optimization methods (other than the hypervolume-based method) with a primary objective to prune the Pareto front. Therefore, this aspect could open up considerable opportunities for future research to employ different diversity optimization approaches such as Omni-aiNet (Coelho & Von Zuben, 2006) as a pruning method. Readers are suggested to refer to Basto Fernandes, Yevseyeva, Deutz, and Emmerich (2017) to examine other potential diversity optimization methods. Moreover, we also recommend that future researchers conduct a comparative study showcase the benefits of their proposed method over the existing diversity-based methods.
- Ensemble pruning methods:** From the discussion given in the previous section, we can see that even pruning methods within the same group can lead to different pruned sets. Each method is shown to produce superior results when considering its directly related metrics. For instance, in Petchrompo et al. (2022), REI indicated that the K -means method outperformed the K -medoids method, while RCI showed the opposite. The authors also mentioned that the difference in such performance is even more significant as the number of objectives increases. This trait opens up an opportunity for future research to develop an ensemble pruning method that results in a

compromise pruned set. In machine learning, ensemble methods are defined as learning algorithms that train and combine multiple base models to solve a computational problem (Dietterich, 2000; Zhou, 2012). These methods are known to produce better results than a single model because it can greatly reduce the likelihood of delivering extremely poor performance in different indicators. In the context of Pareto pruning, we suggest that future research combines different pruning methods within the same class to obtain the pruned set that satisfies different performance metrics. The emphasis should be laid on the combination of multiple partitioning methods or knee-based methods.

- Combination of different pruning methods:** The findings from this review paper indicated that there is only a paucity of research that combine methods across different classes to identify the pruned set. Since different pruning methods have been developed for different purposes, previous researchers generally focused on improving the performance of methods within a particular class. However, in reality, the DM could exploit different types of methods to arrive at the definitive decision. As a guideline, researchers are suggested to combine a diversity-based with an efficiency-based pruning method. This combination would allow the DM to learn about the Pareto front through well-diversified representative solutions before he/she can choose the most efficient solution from a selected group of interest. Another interesting avenue is to combine an efficiency-based with a preference-based pruning method, so that the DM can identify the definitive solution from multiple knees with competitive efficiency values. This suggestion is also in line with the concept of the preferred global knee solution suggested by Zhang et al. (2021).
- Pruning methods for stochastic problems:** The vast majority of pruning methods discussed in this paper are based on the deterministic setting of the Pareto front. Only a few studies mentioned in Section 6 have attempted to incorporate uncertainties into their models by using the solution robustness as their pruning instructions. However, their objective is still the same as in other existing pruning methods, that is to identify promising solutions from a deterministic Pareto front. Recent MOO studies have made great strides in extending the notions of Pareto optimality under uncertainty (Engau & Sigler, 2020; Selcuklu, Coit, & Felder, 2020). A representative example of such extensions can be found in the p -Pareto set proposed by Selcuklu et al. (2020). In this concept, a dominated solution in the deterministic problem may be included in the p -Pareto set if a non-dominated solution in the same region has a high level of uncertainty. In other words, there is a possibility that the solution that is found to be dominated in the deterministic space could become non-dominated in the stochastic setting. As more solutions are included in the stochastic Pareto front, the DM could be faced with an even more exacting task in choosing the final solution. Therefore, we suggest that future studies focus on developing a stochastic pruning method to address this issue.
- Performance indicators for stochastic pruned set:** Following the recommendation on the stochastic pruning method, we propose that future researchers also develop performance indicators for evaluating the stochastic pruned set. We can put forward two main recommendations for future research. First, to help the DM in comprehending extended Pareto sets in a stochastic setting, a customized diversity indicator should be developed. This indicator should take into account not only the distribution of solutions in the deterministic equivalent objective space but also the level of uncertainty associated therewith. Second, another indicator that can measure how well a solution is aligned with the DM's risk preference should be implemented. This indicator is expected to help the DM identify a focused set of representative solutions in case the DM can precisely express his/her attitude towards risk prior to the resolution process. We recommend that future studies explore the possibility of integrating the existing indicators discussed in Section 7 with the

Pareto uncertainty index (Selcuklu et al., 2020) or the robustness space (Jornada & Leon, 2016).

- **Review paper for multi-objective optimization in machine learning:** Having investigated existing review papers and surveys, we have ascertained different applications of MOO methods in various fields. In the past few years, it is evident that researchers and practitioners have made impressive strides in applying MOO to machine learning (ML). Thus, an updated review paper that compiles and analyzes studies that hybridize MOO with ML should be conducted. As a guideline, we recommend that such review paper should engage with the following topics. At the pre-processing stage, the application of MOO in feature selection, missing data imputation, and discretization should be discussed. During learning tasks, MOO can be used for the optimization of the parameters of supervised ML algorithms. A common use here would include situations where specific algorithmic parameters are fine-tuned in order to maximize predictive performance. The use of multi-objective algorithms here has focussed on balancing between algorithmic performance indicators, especially when uncertainty is explored (Pearce, Zaki, Brintrup, & Neely, 2018; Fathy, Jaber, & Brintrup, 2021). Moreover, in unsupervised learning, MOO is often used for optimizing performance outcomes in clustering (Mukhopadhyay, Maulik, & Bandyopadhyay, 2015) and association rule mining (Minaei-Bidgoli, Barmaki, & Nasiri, 2013). More recently, researchers have also investigated the development of multi-objective methods for reinforcement learning. In this respect, multi-objective reinforcement learning has been proposed as a generalization of standard reinforcement learning where the scalar reward signal is extended to multiple feedback signals, so that a set of Pareto efficient policies are learned by agents in a single run (Moffaert & Nowé, 2014).

9. Conclusions

This article reviewed the literature on MOO approaches with an emphasis on the pruning method. Many industrial engineering problems involve the consideration of multiple criteria that conflict with each other. Traditional MOO methods either convert multiple criteria into a single-objective function (*a priori*) or generate a Pareto optimal set to determine the trade-off of conflicting objectives (*a posteriori*), while more recent methods (interactive) require the DM to continuously interact with the algorithm to search for his/her preferred solutions. Nonetheless, the *a priori* method could result in an ineffectual solution if the DM's preference cannot be articulated, whereas the Pareto optimal set obtained from the *a posteriori* method usually contains too many solutions. As an alternative to the interactive method, which is inherently time-consuming, later researchers develop the pruning method to identify a subset of Pareto optimal solutions that is more comprehensible to the DM. In the pruning method, the selection of the representative solutions is based on a predefined instruction, and its resolution process is mostly independent of the DM.

There have been numerous studies devoted to the development of the pruning method. However, previous researchers adopted different instructions in the pruning process and sometimes employed different terms to refer to their specific methods. To distinguish the pruning method from other MOO methods, this article established a classification scheme of MOO methods based on the resolution process, the final output, and the role of the DM. With an aim to facilitate readers in their literature search, we further categorized the pruning method into four classes: preference-based, diversity-based, efficiency-based, and problem-specific methods. Besides discussing specific methods within each class, we also compiled the performance indicators employed by previous researchers to assess the pruned set.

Having delved into the state-of-the-art research in the field, we conducted a comparative analysis to examine the characteristics of different pruning methods to identify their advantages and disadvantages. This discussion allowed us to not only put forward suggestions on

conductive methods for different decision-making situations but also identify important gaps in current research. Ultimately, we made five broad streams of recommendations. First, we suggested that future academics develop novel pre- and intra-optimality pruning methods to address many-objective problems. Second, we recommended that the application of alternative diversity optimization approaches in Pareto pruning is investigated. Third, we found that there is still room for improvement in achieving a pruned set that better suits the DM's requirements if multiple methods are combined. Fourth, we suggested that future research extends the existing pruning method to deal with the Pareto optimal set with uncertainties. This extension will allow the DM to exploit the knowledge about the uncertainty to identify promising solutions in stochastic objective space. Finally, we suggested that an updated review paper emphasizing on the hybridization of MOO and ML is conducted.

CRedit authorship contribution statement

Sanyapong Petchrompo: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Writing – review & editing. **David W. Coit:** Conceptualization, Writing – review & editing, Formal analysis. **Alexandra Brintrup:** Conceptualization, Writing – review & editing, Investigation. **Anupong Wannakrairot:** Conceptualization, Visualization. **Ajith Kumar Parlikad:** Conceptualization, Methodology, Writing – review & editing, Supervision.

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